

BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING

**Adapting Spectral Learning for Any-Resolution AI-Generated Medical
Image Detection**

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Declaration of Candidate

This is to certify that the work presented in this thesis is the outcome of the analysis and experiments carried out by **Kashshaf Labib**, **Navid Kamal**, and **Mir Sayad** under the supervision of **Dr. Md. Hasanul Kabir**, Professor & Head, Department of Computer Science and Engineering and co-supervision of **Dr. Rafsanjany Kushol**, Assistant Professor, Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Bangladesh. It is also declared that neither this thesis nor any part of it has been submitted anywhere else for any degree or diploma. Information derived from the published and unpublished work of others have been acknowledged in the text and a list of references is given.

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Dedicated to our families

Contents

Declaration of Candidate	iii
Dedication	iv
List of Figures	ix
List of Tables	xi
List of Abbreviations	xii
Acknowledgement	xiii
Abstract	xiv
1 Introduction	1
1.1 Background and Motivation	1
1.2 AI-Generated Image Detection: State of the Art	2
1.2.1 SPAI: Spectral AI-Generated Image Detection	3
1.3 The Medical Domain Gap	4
1.3.1 Spectral Statistics of Medical Imaging	4
1.3.2 Physics of Acquisition Noise in Medical Imaging	6
1.3.3 Backbone Mismatch	6
1.4 Problem Statement	7
1.4.1 The Domain Gap Problem	7
1.4.2 The Evaluation Gap Problem	7
1.5 Research Questions	8
1.6 Summary of Contributions	8
1.7 Thesis Organisation	9
2 Related Work	10

2.1	CNN-Based Fake Image Detection and Cross-Generator Generalisation	10
2.2	Frequency-Domain Analysis of GAN-Generated Images	12
2.2.1	Frequency Artifacts in GAN Architectures	12
2.2.2	Upsampling as the Root Cause	13
2.2.3	Frequency-Based Detection	14
2.2.4	Robustness and Limitations	15
2.3	Neighbouring Pixel Relationships for Generalizable Deepfake Detection	15
2.3.1	From Global Frequency to Local Spatial Artifacts	16
2.3.2	The NPR Representation	16
2.3.3	Experimental Results and Relevance to Medical Detection . .	17
2.4	Zero-Shot Detection via Lossless Coding Entropy	18
2.4.1	Motivation and Conceptual Basis	19
2.4.2	Experimental Results	19
2.4.3	Limitations and Connection to Medical Detection	20
2.5	Masked Frequency Modeling for Self-Supervised Pre-Training	21
2.5.1	Motivation: Spatial Redundancy in Masked Image Modeling .	21
2.5.2	The MFM Pre-Training Pipeline	22
2.5.3	Key Empirical Findings	23
2.5.4	Relation to Low-Level Image Restoration	24
2.5.5	Connection to SPAI and the Medical Domain Limitation . . .	24
2.6	SPAI: Any-Resolution AI-Generated Image Detection by Spectral Learning	25
2.6.1	Core Principle: Invariance of the Real Spectral Distribution . .	25
2.6.2	Masked Spectral Learning	26
2.6.3	Spectral Reconstruction Similarity	27
2.6.4	Spectral Context Vector	28
2.6.5	Spectral Context Attention	29
2.6.6	State-of-the-Art Performance and Robustness	30
2.6.7	Why SPAI Fails on Medical Images	30
2.7	Convolutional Vision Transformer (CvT)	31
2.7.1	Motivation: Limitations of Pure Vision Transformers	31
2.7.2	Architectural Design	32
2.7.3	Model Variants and Efficiency	33
2.7.4	Relevance to SPAI Backbone Substitution	33
2.8	AI-Generated Medical Image Detection: Domain-Specific Prior Work	34
2.8.1	CT-GAN: The Medical Manipulation Threat	34
2.8.2	M3DSynth: A Benchmark for CT Manipulation Detection . . .	35

2.8.3	AFFETDS: Adversarial Feature Fusion for MRI Tumour Manipulation Detection	36
2.8.4	DSKI and MedForensics: Towards Multi-Modal Medical Forensics	37
3	Proposed Methodology	39
3.1	Backbone Replacement: ViT-B/16 to CvT-13	39
3.2	Medical MFM Pretraining	40
3.3	Physics-Informed Frequency Masking	40
3.4	Medical AI-Generated Image Dataset Pipeline	41
3.4.1	Stage 1: Report Parsing and Filtering	41
3.4.2	Stage 2: Structured Prompt Extraction via LLM	42
3.4.3	Stage 3: Multi-View Detection and Prompt Splitting	43
3.4.4	Stage 4: Image Prompt Formatting	44
3.4.5	Stage 5: Image Generation	44
3.4.6	Aspect Ratio Preservation	45
3.4.7	Output Format and Provenance Tracking	45
4	Results and Discussion	46
4.1	Datasets and Experimental Setup	46
4.1.1	Datasets	46
4.1.2	Experimental Configuration	47
4.1.3	Evaluation Metrics	47
4.2	Experiment 1: Spectral Domain Gap in Medical Images	47
4.3	Experiment 2: Controlled Backbone Comparison	48
4.3.1	Implementation Details	48
4.3.2	Training Dynamics	49
4.3.3	Per-Generator Generalisation	50
4.3.4	Aggregate Efficiency Comparison	51
4.3.5	Interpretation of Results Relative to Published SPAI	51
4.4	Experiment 3: Medical Image Dataset & Generation Pipeline	52
4.4.1	Source Dataset	52
4.4.2	Generation Pipeline	52
4.4.3	Dataset Composition	52
4.4.4	Modality Breakdown (Generated)	53
4.4.5	Implementation Results	53
4.5	Challenges and Limitations	54
4.5.1	Ethical Considerations	54

4.5.2	Implications of Findings	55
4.5.3	Chapter Summary	56
5	Conclusion	57
	References	59

List of Figures

2.1	Average frequency spectra of real and CNN-generated images across 11 generation models, from Wang et al. [45]. High-pass filtered images reveal periodic spectral artifacts (visible as dots or grid lines) in most synthetic images, while real image spectra are comparatively smooth. These patterns, particularly absent in BigGAN and DeepFake outputs, provide evidence that CNN upsampling architectures introduce systematic, detectable frequency-domain fingerprints.	11
2.2	Mean DCT spectra of images generated by four GAN architectures (BigGAN, ProGAN, StyleGAN, SN-DCGAN) trained on the Stanford Dogs dataset, alongside the real training data spectrum (from Frank et al. [21]). The real spectrum (leftmost) exhibits a smooth $1/f^\alpha$ falloff. All GAN spectra display elevated high-frequency energy and a characteristic grid-like artifact pattern, whose severity varies by architecture but is present universally. These artifacts are caused by the upsampling operations that all current GAN generators employ to map from a compact latent space to the full image resolution.	13
2.3	Overview of the MFM pre-training pipeline (from Xie et al. [48]). Each input image is converted to the frequency domain via 2D FFT. A low-pass or high-pass circular mask is applied to the frequency spectrum, and the inverse FFT produces a spatially corrupted image that is fed to the encoder. A lightweight linear decoder then predicts the masked frequency values on the spectrum using a frequency reconstruction loss. Because no mask tokens are inserted, the same pipeline applies to both ViT and CNN encoders without architectural modification. . .	22

2.4	Overview of the SPAI architecture (from Karageorgiou et al. [27]). A MFM-pretrained ViT-B/16 backbone (kept frozen) encodes the spectral distribution of real images. Spectral Reconstruction Similarity (SRS) measures the divergence between representations of an image and its frequency-filtered components at each transformer block. The Spectral Context Vector (SCV) captures the spectral context under which SRS values are computed. Spectral Context Attention (SCA) fuses patch-level spectral vectors into an image-level representation with $\mathcal{O}(K)$ complexity, enabling any-resolution processing. A three-layer MLP produces the final binary prediction.	26
2.5	Overview of the CvT architecture (from Wu et al. [47]). (a) The hierarchical multi-stage pipeline: each stage begins with a Convolutional Token Embedding layer that performs an overlapping strided convolution on the 2D-reshaped token map, progressively reducing spatial resolution while increasing feature dimension. (b) The Convolutional Transformer Block: the standard position-wise linear projection for queries, keys, and values is replaced with a depth-wise separable Convolutional Projection, enabling local spatial context modelling within the attention mechanism. The classification token is appended only in the final stage.	32
3.1	Overview of the synthetic radiology image generation pipeline. Solid arrows denote the primary data flow; dashed arrows indicate optional multimodal inputs used when reference images are available. Orange boxes are processing stages; blue cylinders are data sources/sinks; red boxes are generative backends; diamond nodes are routing decisions. .	42

List of Tables

1.1	Spectral slope statistics for natural and medical image domains. Data collected at 256×256 grayscale resolution. Natural real: COCO 2017 validation set. Natural synthetic: CIFAKE. Medical real: Indiana University Chest X-rays. Medical synthetic: LLM-guided DALL-E 3 generated X-rays.	5
2.1	Source identification accuracy of classifiers trained on raw pixels versus DCT coefficients, reproduced from Frank et al. [21]. The frequency domain consistently improves all classifiers, with the largest gains for simpler methods. The shallow CNN trained on DCT coefficients matches the large CNN of Yu et al. at approximately 1.9% of its parameter count.	14
2.2	Mean detection accuracy across 28 generative models (38 sub-test sets), reproduced from Tan et al. [44]. All detectors are trained on ProGAN only. NPR outperforms all baselines by at least 11.6 percentage points, demonstrating substantially stronger cross-source generalisation. . . .	17
2.3	Evaluation of different pre-training backbones within SPAI, reproduced from Karageorgiou et al. [27]. Average AUC over 5 sources of real images and 13 generative models is reported. The MFM-pretrained ViT-B/16 outperforms both CLIP and DINOv2 despite being pre-trained on $117\times$ and $333\times$ less data respectively.	27
2.4	Comparison of AI-generated image detectors across 13 generative models, reproduced from Karageorgiou et al. [27]. Average AUC over 5 sources of real images is reported. SPAI achieves the highest average AUC across all image sizes, with a 5.5% absolute improvement over the prior state-of-the-art RINE [30].	30

3.1	Structured attributes extracted from free-text radiology reports by the LLM extraction stage. The impression field is intentionally omitted from downstream image prompts to prevent generation models from rendering diagnostic text within the synthetic radiograph.	43
3.2	Named sections of the formatted image generation prompt. Sections are conditionally included based on data availability.	44
4.1	Training configuration used in pilot and target full experiments. . . .	47
4.2	Spectral slope α comparison between real and synthetic images across natural and medical domains. The $ \Delta\alpha $ detection signal is $4.5\times$ smaller in the medical domain, directly explaining SPAI’s near-chance zero-shot accuracy on clinical data.	48
4.3	CvT-13 per-stage output shapes and adapter configuration. All stages are projected to the uniform $(B, 196, 768)$ interface required by SPAI’s SRS computation.	49
4.4	Pilot training dynamics of ViT-B/16 and CvT-13 variants.	49
4.5	Per-generator detection performance across 17 unseen generators. Both variants were trained on Latent Diffusion, Stable Diffusion, DDPM, and Palette only; all evaluation generators are entirely unseen during training. $\Delta\text{AUC} = \text{CvT} - \text{ViT}$	50
4.6	Aggregate efficiency comparison between original SPAI and the proposed variants. CvT-13 matches ViT-B/16-level generalisation at 77% fewer backbone parameters, trained on $21\times$ less data in $11.7\times$ fewer epochs.	51
4.7	Distribution of generated images across modalities.	53
4.8	Detection accuracy by image source on the medical dataset.	53

List of Abbreviations

AID	AI-Generated Image Detection
AUC	Area Under the ROC Curve
CNN	Convolutional Neural Network
CT	Computed Tomography
CvT	Convolutional Vision Transformer
DM	Diffusion Model
FFT	Fast Fourier Transform / 2D Discrete Fourier Transform
GAN	Generative Adversarial Network
LLM	Large Language Model
MFM	Masked Frequency Modeling
MIM	Masked Image Modeling
MLP	Multi-Layer Perceptron
MRI	Magnetic Resonance Imaging
MSE	Mean Squared Error
PSD	Power Spectral Density
SCA	Spectral Context Attention
SCV	Spectral Context Vector
SPAI	SPECTral AI-Generated Image Detection
SRS	Spectral Reconstruction Similarity
SSL	Self-Supervised Learning
ViT	Vision Transformer

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Abstract

AI-generated medical images pose a growing risk to clinical practice by enabling the fabrication of diagnostic evidence, contaminating training data for medical AI systems, and facilitating fraud in clinical studies. Although state-of-the-art (SOTA) spectral-based methods have shown strong performance in detecting synthetic images in natural photography, their effectiveness drops significantly in the medical domain. In our study, these methods achieve only marginally better-than-random performance on a newly constructed medical image dataset.

Through detailed analysis, we find that the difference between real and synthetic medical images in the frequency domain is much less pronounced than in natural images, leaving existing approaches with limited useful signal for reliable detection.

To address this limitation, we propose three key improvements. First, we replace the standard transformer backbone with a hybrid CNN-transformer architecture that captures local spatial patterns and supports efficient learning from smaller datasets. Second, we introduce a physics-informed frequency masking strategy that adapts to different imaging modalities, reducing the influence of inherent acquisition noise such as CT noise and MRI artifacts. Third, we develop a new medical image dataset using a controlled generation pipeline with multiple external image generators to ensure robust and unbiased evaluation.

In a pilot study, our approach demonstrates strong performance across a diverse set of unseen generators, achieving high detection accuracy while using significantly fewer parameters, less training data, and fewer training iterations. These results highlight the effectiveness of our method and its potential as a data-efficient solution for detecting AI-generated medical images across different modalities and resolutions.

Chapter 1

Introduction

1.1 Background and Motivation

The rapid advancement of generative artificial intelligence has fundamentally altered the landscape of image synthesis. Early generative approaches, primarily based on Generative Adversarial Networks (GANs) [22], established the capability to produce photorealistic visual content from random noise. More recently, Diffusion Models [25], [39] have set new standards in image fidelity, capable of generating high-resolution imagery conditioned on arbitrary textual or visual inputs. The continued availability of powerful open-source and commercial platforms, including Stable Diffusion, DALL-E 3, Midjourney, and Gemini, has made high-quality synthetic image generation accessible to virtually anyone.

While these advances carry enormous creative and scientific potential, they introduce a particularly grave threat in a domain where visual authenticity is foundational: *medical imaging*. Synthetic chest X-rays, CT slices, MRI scans, and histopathology slides produced by modern generative models are now visually indistinguishable from authentic clinical scans, even under the scrutiny of trained radiologists. This visual indistinguishability opens critical attack surfaces across the healthcare ecosystem:

- Fabrication of imaging evidence in clinical drug trials;
- Corruption of training datasets used to build AI diagnostic systems;
- Generation of fraudulent insurance records and medical documentation;
- Introduction of misleading public health surveillance data.

Unlike natural photograph forgery, where manipulations may be identified through

semantic inconsistencies, extra fingers, incorrect shadow geometry, perspective errors, the consequences of undetected medical image forgery extend directly to patient safety and institutional trust in clinical science. The threat is not merely theoretical. As Dufour et al. [15] documented, synthetic imagery is already proliferating online in harmful contexts, blurring the provenance between real and generated content. In medical practice, where imaging data underpins diagnoses, regulatory decisions, and the training of clinical AI, the integrity of visual content is a public health imperative.

1.2 AI-Generated Image Detection: State of the Art

The computer vision and image forensics communities have proposed a substantial body of AI-Generated Image Detection (AID) methods. These approaches broadly fall into three paradigms.

Semantic inconsistency detection exploits structural artifacts in facial geometry [4], perspective [19], shadow geometry [41], and lighting [18] that current generative models fail to render consistently. While effective against early GAN architectures, these methods are rapidly becoming obsolete as state-of-the-art generators, Stable Diffusion 3, Midjourney v6.1, have largely eliminated such gross geometric inconsistencies.

Low-level artifact capture operates in either the spatial domain [9], [45] or the spectral domain [16], [21], exploiting subtle pixel-level traces or frequency-domain anomalies introduced by the generation process. Generative models are known to over-smooth image content, producing a characteristic upward shift in the power spectral density (PSD) slope α relative to real images. Frequency-domain detectors exploit this spectral discrepancy as a detection signal. However, as different generative architectures introduce significantly different artifacts [2], these detectors generalise poorly to generators unseen during training.

Pre-trained encoder features represent the most recent direction, leveraging representations from large vision-language models such as CLIP [38] to capture both semantic and low-level artifacts within a unified framework [10], [30]. RINE [30], the prior state-of-the-art before SPAI, achieves 85.5% average AUC across diverse generators using an intermediate-layer representation approach on a CLIP backbone pre-trained on 400 million images.

Despite this progress, a fundamental limitation persists across all paradigms: detectors trained to capture the artifacts of specific generators fail to generalise when evaluated on generators with different characteristics. This limitation is structurally in-

tractable if detection relies on learned artifact signatures, because the space of possible generators is open-ended and continuously expanding.

1.2.1 SPAI: Spectral AI-Generated Image Detection

Karageorgiou et al. [27] (CVPR 2025) proposed a fundamentally different paradigm: rather than learning to recognise the artifacts of specific generators, learn a spectral model of the real image distribution and detect AI-generated images as out-of-distribution samples of that model. This approach, *Spectral AI-Generated Image Detection* (SPAI), rests on the observation that the spectral distribution of real images constitutes an *invariant* pattern. It is determined by long-term technological factors such as optics, sensors, and image processing pipelines, rather than by the introduction of any specific generative model. AI-generated images, by contrast, are out-of-distribution with respect to this invariant spectral model, regardless of the specific generator used.

SPAI operationalises this principle through three components. First, *Masked Spectral Modeling* (MFM) uses a Masked Frequency Modeling pretrained ViT-B/16 backbone [48], kept frozen throughout downstream training, to encode the spectral distribution of real images. The backbone is pretrained via a frequency reconstruction pretext task: the 2D Fourier spectrum is split into low- and high-frequency bands via a circular mask at radius $r = 16$, and the model learns to reconstruct the masked band from the complementary one. Second, *Spectral Reconstruction Similarity* (SRS) measures the alignment between the representations of an image and its low-pass and high-pass frequency components at each of the ViT’s $N = 12$ transformer blocks. Real images, being in-distribution for the spectral model, yield mutually consistent cross-band representations, while AI-generated images do not. Third, *Spectral Context Attention* (SCA) enables any-resolution processing by splitting large images into patches of size 224×224 and fusing their spectral vectors through a learned cross-attention mechanism with $\mathcal{O}(K)$ complexity, where K is the number of patches. A *Spectral Context Vector* (SCV) additionally encodes what kind of spectral content each patch contains, preventing naturally smooth images from being incorrectly flagged as AI-generated.

Together, these components achieve 91.0% average AUC across 13 unseen generative models, a 5.5% absolute improvement over the prior state-of-the-art, while using a backbone pretrained on only 1.2 million images, outperforming methods based on CLIP (400 million) and DINOv2 (142 million).

1.3 The Medical Domain Gap

Despite SPAI’s strong performance on natural image benchmarks [27], we find that it fails critically when applied to the medical imaging domain. Our zero-shot evaluation on a dataset of real chest X-rays (Indiana University Chest X-ray Collection) paired with LLM-guided synthetic X-rays generated by third-party models achieves only 61.3% average accuracy, barely above the 50% random baseline.

1.3.1 Spectral Statistics of Medical Imaging

The root cause of this failure is spectral. Every image decomposes via 2D Discrete Fourier Transform (DFT) into its constituent spatial frequency components, separating broad low-frequency structures (e.g. smooth tissue regions) from fine high-frequency details (e.g. sharp edges and noise). The *Power Spectral Density* (PSD) describes how image energy is distributed across these frequencies. For natural images, the radially averaged PSD follows a well-established power law [20]:

$$P(f) \propto f^{-\alpha}, \quad (1.1)$$

where f denotes spatial frequency, and the *spectral slope* α quantifies the rate at which energy falls off with increasing frequency. A higher α indicates a steeper falloff: fewer high-frequency components, resulting in a smoother image overall. On a log–log plot, this relationship appears as a straight line with slope $-\alpha$, and the *goodness-of-fit* R^2 measures how well the data conform to this power-law model ($R^2 = 1$ is a perfect fit).

AI-generated images are characteristically over-smooth, producing systematically higher α values than their real counterparts [16], [21]. This arises from architectural properties of generative models: upsampling operations such as transposed convolution fail to reproduce the high-frequency energy present in real images, leaving generated images deficient in fine spatial detail [16], [51].

Table 1.1 summarises our analysis of 150 real images and 150 AI-generated images from both the natural and medical domains.

Four findings emerge directly from this data.

Finding 1: The detection gap is 4.5× smaller in the medical domain. The alpha gap $|\Delta\alpha|$ falls from 0.775 in natural images to 0.170 in medical images, only 22% of the natural image gap. This directly compresses the primary detection signal that frequency-based detectors exploit [16], [27].

Table 1.1: Spectral slope statistics for natural and medical image domains. Data collected at 256×256 grayscale resolution. Natural real: COCO 2017 validation set. Natural synthetic: CIFAKE. Medical real: Indiana University Chest X-rays. Medical synthetic: LLM-guided DALL-E 3 generated X-rays.

Group	$\bar{\alpha}$	σ_{α}	R^2	$ \Delta\alpha $
Real Natural	2.556	0.288	0.988	0.775
Synthetic Natural	3.331	0.244	0.956	
Real Medical	2.868	0.183	0.994	0.170
Synthetic Medical	3.038	0.133	0.996	

Finding 2: Medical images are spectrally homogeneous. The standard deviation of α for real medical images ($\sigma_{\alpha} = 0.183$) is 37% lower than for natural images ($\sigma_{\alpha} = 0.288$). Both real and synthetic medical images cluster in a narrow spectral band with only $|\Delta\alpha| = 0.17$ between them, leaving almost no room for a decision boundary. This low spectral variance reflects the inherently uniform structural composition of radiographic images, which approximate the $1/f$ power-law model more closely than natural photographs [20], [33].

Finding 3: Real medical images already resemble AI-generated natural images spectrally. Real chest X-rays have $\bar{\alpha} = 2.87$, approaching the $\bar{\alpha} \approx 3.33$ profile of AI-generated natural images. This is consistent with the smooth, low-noise nature of X-ray acquisition physics, where beam filtering and projection averaging suppress high-frequency content [32], [33]. As a result, SPAI’s internal model of “real image spectral distribution”, calibrated on natural-scene data with $\alpha \approx 2.5$, is poorly matched to clinical data.

Finding 4: Statistical significance does not imply practical discriminability. To quantify distributional separability between real and synthetic images, we employ two complementary statistical measures. The *two-sample Kolmogorov–Smirnov (KS) test* [29], [42] is a non-parametric test that computes the maximum absolute difference between the empirical cumulative distribution functions (ECDFs) of two groups; a KS statistic near 1 indicates near-complete separation, while values near 0 indicate heavy overlap. *Cohen’s d* [8] measures the practical magnitude of this difference in units of pooled standard deviation:

$$d = \frac{|\mu_A - \mu_B|}{\sigma_{\text{pooled}}}, \quad (1.2)$$

with conventional thresholds of $d < 0.2$ (negligible), $0.2\text{--}0.5$ (small), $0.5\text{--}0.8$ (medium), and $d > 0.8$ (large) [8].

The KS statistic for medical images ($\text{KS} = 0.51$, $p = 6.93 \times 10^{-18}$, Cohen’s $d = 1.07$)

is statistically significant, but Cohen’s d is $2.7\times$ weaker than the natural image counterpart (Cohen’s $d = 2.91$, $KS = 0.87$). This illustrates a critical distinction: statistical significance (driven by sample size) does not guarantee practical discriminability. The distributions overlap substantially in the medical domain, and the compressed $|\Delta\alpha|$ leaves a spectral detector with far less margin to separate real from synthetic images [21], [45].

1.3.2 Physics of Acquisition Noise in Medical Imaging

This spectral homogeneity is a fundamental consequence of the physics of medical image acquisition, and it introduces a second, more insidious failure mode.

CT imaging is dominated by Poisson photon noise in the high-frequency band [33]: real CT images contain structured high-frequency content arising from X-ray photon statistics, which is absent in AI-generated counterparts.

MRI imaging contains structured k-space aliasing artifacts from the acquisition protocol [32], [37]: real MRI images exhibit characteristic high-frequency structured noise from the sampling trajectory, which AI-generated images do not reproduce.

X-ray imaging carries detector and photon scatter noise: the specific noise profile of the digital detector is present in every real scan.

A fixed frequency mask, such as the circular mask at radius $r = 16$ used by SPAI, which covers only 1.6% of all frequency bins, cannot distinguish between these real-image acquisition noise structures and the absence of high-frequency content in AI-generated medical images. The result is that SPAI’s fixed mask actively confuses real imaging physics with AI generation artifacts, producing false negatives on real images and misattributing acquisition noise as evidence of synthesis.

1.3.3 Backbone Mismatch

A further failure mode arises from SPAI’s backbone. The MFM-pretrained ViT-B/16 was trained on ImageNet, a natural image dataset with $\bar{\alpha} \approx 2.5$. Its spectral prior, the learned model of what “real image frequencies” look like, is calibrated to natural photograph statistics and not to the substantially different spectral profiles of clinical modalities. Retraining ViT-B/16 via MFM on medical data would require 1.2–14 million medical images [9], a scale far beyond what is available in curated clinical repositories, which typically contain tens to hundreds of thousands of scans. Furthermore, the ViT-B/16 architecture offers no local spatial inductive bias, requiring all

spatial relationships to be learned from data, making it data-hungry by design.

1.4 Problem Statement

AI-generated medical images pose serious and escalating risks across the healthcare ecosystem. The fabrication of clinical trial evidence using synthetic scans can corrupt drug approval processes; misleading synthetic training data can compromise the reliability of AI diagnostic systems; and fraudulent medical records can be exploited for insurance fraud and identity theft. These threats are not theoretical, the same generative models that produce convincingly realistic imagery for entertainment and design have been shown to produce clinically plausible radiology reports and medical scans.

1.4.1 The Domain Gap Problem

Existing AI-generated image detectors are designed and validated on natural photographs. Medical images, however, have fundamentally different spectral and statistical characteristics:

- Smooth gradients with high spectral slopes ($\alpha \geq 2.87$), approaching the spectral profile of AI-generated natural images;
- Acquisition-specific noise patterns: Poisson noise in CT, k-space aliasing artifacts in MRI, detector and scatter noise in X-ray;
- Narrow spectral variance (standard deviation of $\alpha = 0.183$) that clusters real and synthetic medical images in overlapping spectral bands.

Our quantitative spectral analysis demonstrates that the alpha gap between real and synthetic medical X-rays ($|\Delta\alpha| = 0.170$) is only 22% of the corresponding gap for natural images ($|\Delta\alpha| = 0.775$). This 4.5 $\times$ reduction in discriminative spectral signal directly causes frequency-based detectors like SPAI to fail on clinical data, with zero-shot accuracy falling to 61.3%, marginally above random chance.

1.4.2 The Evaluation Gap Problem

Existing synthetic medical image datasets rely almost exclusively on Stable Diffusion variants. This creates a circular evaluation problem: detectors trained and tested against Stable Diffusion artifacts cannot be claimed to generalize to the broader landscape of generative models. Real-world forensic scenarios involve images produced

by DALL-E 3, Midjourney v6, Gemini, and other third-party systems with distinct spectral signatures.

1.5 Research Questions

This thesis is guided by the following primary research question:

Can spectral learning be adapted to reliably detect AI-generated medical images across modalities and resolutions, using a data-efficient, physics-aware architecture?

This decomposes into three sub-questions:

1. What is the spectral basis for the failure of existing frequency-domain detectors (specifically SPAI) on medical images, and can it be quantified rigorously using PSD metrics?
2. Does replacing the ViT-B/16 backbone with the data-efficient CvT-13 maintain or improve detection performance across unseen generators, while significantly reducing parameter count and training data requirements?
3. Does modality-conditioned physics-informed frequency masking improve discriminative spectral signal and detection accuracy compared to the fixed circular mask, particularly for CT and MRI modalities?

1.6 Summary of Contributions

The principal contributions of this thesis are:

1. **Spectral Domain Analysis.** A rigorous quantitative characterisation of the spectral domain gap in medical imaging, using power spectral density slope α , Kolmogorov-Smirnov statistics, and Cohen’s d effect size, across 150 real and 150 AI-generated chest X-rays and natural images. We demonstrate that the $|\Delta\alpha|$ detection signal is 4.5× smaller in medical images, directly explaining SPAI’s near-chance performance.
2. **CvT-13 Backbone Adaptation.** A systematic architectural substitution of SPAI’s ViT-B/16 backbone with CvT-13, including the design of a `SpatialChannelAdapter` module for bridging heterogeneous multi-stage CvT feature maps to SPAI’s downstream components. Pilot experiments demonstrate 97.2% average AUC across

17 unseen generators, matching the ViT-B/16 baseline at 77% fewer backbone parameters (20 M vs. 85.8 M), trained on $21\times$ less data in $11.7\times$ fewer epochs.

3. **Physics-Informed Frequency Masking.** A learned, modality-conditioned frequency mask that suppresses acquisition-physics noise bands in CT, MRI, and X-ray imagery, preventing misclassification of real imaging phenomena as AI-generated artifacts.
4. **Novel Medical AI-Generated Image Dataset.** A dataset pipeline employing LLM-guided prompt generation and diverse third-party generators (DALL-E 3, Midjourney v6, Gemini), producing the first synthetic medical image benchmark that explicitly avoids the Stable Diffusion monoculture present in all existing comparable datasets.

1.7 Thesis Organisation

The remainder of this thesis is organised as follows. Chapter 2 reviews the related literature on AI-generated image detection, spectral analysis for forensics, masked image modelling, and synthetic medical image generation. Chapter 3 presents the proposed methodology in full, covering the CvT-13 backbone adaptation, the physics-informed frequency masking mechanism, and the medical dataset pipeline. Chapter 4 describes the experimental setup and presents the results of our pilot study and full evaluation, including per-generator detection performance, ablation studies, and spectral analysis. Chapter 5 concludes the thesis, summarises key findings, acknowledges limitations, and outlines directions for future work.

Chapter 2

Related Work

2.1 CNN-Based Fake Image Detection and Cross-Generator Generalisation

A foundational challenge in AI-generated image detection is determining whether detectors trained to identify fakes from one generative model can transfer to others. Early work framed this as a technique-specific classification problem, training binary classifiers on real images paired with fakes from a single generator and evaluating on that same generator [40]. While such detectors achieved high in-distribution accuracy, they were widely observed to fail when the generator or even the image domain was changed, with cross-model performance often approaching chance [12], [51]. This fragility is expected, given that different architectures, loss functions, and training datasets introduce fundamentally distinct low-level artifact signatures.

Wang et al. [45] challenged this prevailing view in a landmark study that remains directly relevant to our work. They demonstrated that a standard ResNet-50 classifier, trained solely on images from a single GAN (ProGAN [28]) to distinguish real from fake, generalises surprisingly well to images generated by ten other architecturally diverse models, spanning unconditional GANs (StyleGAN, BigGAN), conditional image translation networks (CycleGAN, StarGAN, GauGAN), perceptual-loss models (CRN, IMLE), and low-level vision tasks (super-resolution, seeing-in-the-dark), as well as deepfakes. This finding reframed the detection problem: rather than each new generator requiring its own bespoke detector, there appeared to be systematic, cross-model spectral fingerprints introduced by the upsampling-convolutional structure common to nearly all CNN-based generators.

The authors identified two factors as critical to achieving cross-generator generalisation. First, **data augmentation** via simulated post-processing, specifically, Gaussian blurring and JPEG compression applied stochastically during training, substantially improved generalisation to unseen generators, even when the test images themselves were not post-processed. This augmentation strategy also endowed the detector with robustness to real-world distribution shifts, such as social media compression and re-sizing, which are practically unavoidable in forensic deployment scenarios. Second, **training set diversity**, increasing the number of LSUN semantic categories used for ProGAN training from 2 to 16 classes, produced consistent gains in cross-generator average precision (AP), though with diminishing returns beyond 16 classes. Frequency analysis of the synthesised images revealed periodic spectral patterns in most architectures (Figure 2.1), consistent with aliasing introduced by strided deconvolutional upsampling layers, providing a mechanistic basis for the observed generalisability.

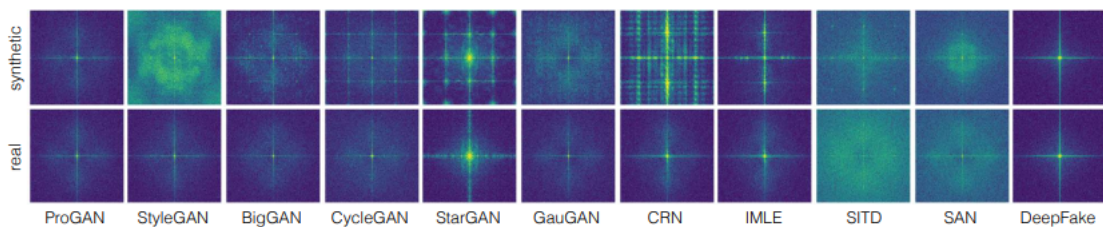


Figure 2.1: Average frequency spectra of real and CNN-generated images across 11 generation models, from Wang et al. [45]. High-pass filtered images reveal periodic spectral artifacts (visible as dots or grid lines) in most synthetic images, while real image spectra are comparatively smooth. These patterns, particularly absent in BigGAN and DeepFake outputs, provide evidence that CNN upsampling architectures introduce systematic, detectable frequency-domain fingerprints.

The study also reported important failure modes that remain pertinent to our medical imaging context. The super-resolution model (SAN) and DeepFake both showed degraded detection when blur augmentation was applied during training, because their discriminative signal resides predominantly in the high-frequency band that blurring destroys. More broadly, the detector’s confidence scores correlated weakly with perceptual image quality across most generators, suggesting the model had learned low-level CNN artifacts rather than high-level realism cues, a property that limits its applicability when these low-level artifacts are absent or suppressed.

Critically, the detector trained by Wang et al. was designed and validated entirely on natural photographs. The spectral fingerprints it exploits, the *gap* between the power spectral density profiles of real and synthetic natural images, are substantially larger in the natural domain than in the medical domain. As we quantify in Chapter 1, the

alpha gap $|\Delta\alpha|$ between real and synthetic images is $4.5\times$ smaller for medical images (0.170) than for natural photographs (0.775). This directly explains why the generalisation that Wang et al. demonstrated on natural images does not transfer to the clinical domain: the common spectral fingerprints that their detector relies upon are largely absent when the generator is applied to medical content, or are masked by modality-specific acquisition noise structures such as Poisson photon noise in CT and k-space aliasing in MRI. Our work builds on the conceptual foundation established by Wang et al., that spectral artifacts are a productive site of forensic analysis, while addressing the domain gap that renders their approach insufficient for medical image detection.

2.2 Frequency-Domain Analysis of GAN-Generated Images

While Wang et al. established that CNN-generated images carry cross-generator spectral fingerprints, Frank et al. [21] provided the first comprehensive *mechanistic* explanation of why these artifacts arise, and demonstrated that frequency-domain representations alone are sufficient for near-perfect detection. Their work is directly foundational to the spectral analysis methodology we employ in this thesis and to our quantitative characterisation of the medical domain gap.

2.2.1 Frequency Artifacts in GAN Architectures

Frank et al. transformed GAN-generated images into the frequency domain using the 2D Discrete Cosine Transform (DCT), which shares the energy compaction and separability properties of the DFT while being computationally efficient for image processing pipelines. For a grayscale image $I \in \mathbb{R}^{N_1 \times N_2}$, the type-II 2D-DCT produces a coefficient matrix $D \in \mathbb{R}^{N_1 \times N_2}$ where each entry D_{k_x, k_y} encodes the contribution of a spatial frequency pair (k_x, k_y) to the image. Low-frequency components dominate the top-left of the DCT heatmap; high-frequency components appear toward the bottom-right.

A core empirical finding of the paper is that the DCT spectra of natural images follow the well-established $1/f^\alpha$ power law, where energy decreases smoothly and gradually from low to high frequencies, while GAN-generated images deviate from this law in two characteristic ways: (i) substantially elevated high-frequency energy in the lower-right quadrant of the spectrum, and (ii) a periodic *grid-like pattern* of spec-

tral peaks, visible across all tested architectures (BigGAN, ProGAN, StyleGAN, SN-DCGAN) and datasets. Figure 2.2 illustrates these spectral artifacts for four GAN architectures trained on the Stanford Dogs dataset.

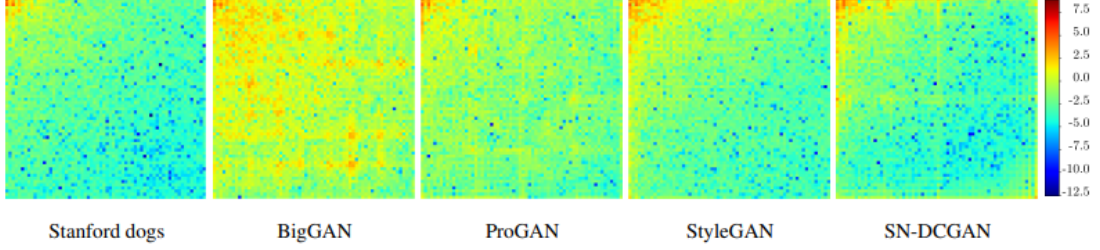


Figure 2.2: Mean DCT spectra of images generated by four GAN architectures (BigGAN, ProGAN, StyleGAN, SN-DCGAN) trained on the Stanford Dogs dataset, alongside the real training data spectrum (from Frank et al. [21]). The real spectrum (left-most) exhibits a smooth $1/f^\alpha$ falloff. All GAN spectra display elevated high-frequency energy and a characteristic grid-like artifact pattern, whose severity varies by architecture but is present universally. These artifacts are caused by the upsampling operations that all current GAN generators employ to map from a compact latent space to the full image resolution.

2.2.2 Upsampling as the Root Cause

The paper identifies the structural origin of these artifacts in the upsampling operations that all GAN generators must employ. A GAN generator defines a mapping from a low-dimensional latent vector to a high-dimensional image, for example, $G : \mathbb{R}^{100} \rightarrow \mathbb{R}^{1024 \times 1024}$ in the StyleGAN instance they analyse. This dimensional-expansion is achieved through successive upsampling stages, each of which must synthesise new pixel values that were not present in the lower-resolution representation.

Frank et al. systematically compare three upsampling strategies, nearest-neighbour, bilinear, and binomial, by training variants of StyleGAN on the LSUN Bedrooms dataset with each strategy substituted in. The DCT spectra confirm a clear gradient: more sophisticated anti-aliasing kernels (larger support, more accurate frequency rejection) progressively reduce the severity of spectral artifacts. Nearest-neighbour upsampling introduces the strongest grid-like patterns; binomial upsampling produces the smoothest spectra. However, artifacts remain detectable even under binomial upsampling, indicating that the structural problem is not merely an implementation choice but an inherent consequence of the low-to-high-dimensional mapping that generative architectures must perform. This finding, that upsampling is the root cause of spectral artifacts in GAN-generated images, is a direct predecessor to our finding that the

acquisition-physics noise structures in medical images (Poisson photon noise in CT, k-space aliasing in MRI) constitute a different kind of structured high-frequency content that fixed frequency masks cannot distinguish from synthesis artifacts.

2.2.3 Frequency-Based Detection

Frank et al. demonstrate that DCT-domain representations dramatically improve the performance of detection classifiers at every level of complexity. For binary detection of StyleGAN faces versus real FFHQ images, a simple linear ridge regression achieves 75.78% accuracy on raw pixels but reaches **100%** accuracy on DCT coefficients, demonstrating complete linear separability in the frequency domain for this task. This is a striking result: images that fool human observers and non-linear image-domain classifiers are trivially separable by a linear classifier once transformed to the frequency domain.

For the harder task of *source identification*, attributing a generated image to the specific GAN architecture that produced it, across four architectures plus real images, the authors show that DCT representations improve all tested classifiers substantially. k-NN accuracy improves by 41–42 percentage points; Eigenfaces by 30–47 points. Their shallow four-layer CNN, trained on DCT coefficients, achieves 99.64% (LSUN) and 99.07% (CelebA) accuracy, matching the much larger CNN of Yu et al., which uses roughly 53× more parameters (~9M vs. ~170K). Table 2.1 reproduces the key accuracy comparison from their source identification experiments.

Table 2.1: Source identification accuracy of classifiers trained on raw pixels versus DCT coefficients, reproduced from Frank et al. [21]. The frequency domain consistently improves all classifiers, with the largest gains for simpler methods. The shallow CNN trained on DCT coefficients matches the large CNN of Yu et al. at approximately 1.9% of its parameter count.

Method	LSUN	Gain	CelebA	Gain
kNN	39.96%		29.22%	
kNN-DCT	81.56%	+41.60%	71.58%	+42.36%
Eigenfaces	47.07%		57.58%	
Eigenfaces-DCT	94.31%	+47.24%	88.39%	+30.81%
CNN (Yu et al.)	98.33%		99.70%	
CNN-DCT (Yu et al.)	99.61%	+1.28%	99.91%	+0.21%
CNN-Pixel	98.95%		97.80%	
CNN-DCT	99.64%	+0.69%	99.07%	+1.27%

2.2.4 Robustness and Limitations

Frank et al. further evaluate the robustness of their DCT-based classifier against five common post-processing perturbations: Gaussian blurring, random cropping, JPEG compression, additive Gaussian noise, and a combination of all four. The DCT classifier outperforms its pixel-domain counterpart on four of the five perturbations when trained on perturbed data, with compression robustness improving by 16.16 percentage points (78.67% vs. 94.83%). The sole exception is additive Gaussian noise, where the pixel-domain classifier retains a marginal advantage, plausibly because Gaussian noise itself introduces high-frequency energy that competes with the spectral artifacts being used as detection signal.

This noise sensitivity is directly relevant to our medical imaging context. Acquisition noise in CT (Poisson photon statistics) and MRI (k-space aliasing) constitutes structured high-frequency content whose spectral signature overlaps with the artifacts that frequency-domain detectors learn to exploit. A fixed circular mask at a single radius, as used by SPAI, cannot distinguish modality-specific acquisition noise from synthesis artifacts, producing the false-negative behaviour we observe in our zero-shot evaluation. Our proposed physics-informed frequency masking addresses this limitation directly by learning a modality-conditioned mask that suppresses known acquisition-noise frequency bands before spectral comparison.

A further acknowledged limitation of Frank et al. is that their classifier can be evaded by white- and black-box adversarial attacks [6], and that more sophisticated upsampling (e.g., binomial anti-aliasing) reduces but does not eliminate detectability. These limitations motivate the SPAI framework’s shift from artifact recognition to spectral out-of-distribution detection, a paradigm that is less contingent on the specific artifact signature of any single upsampling implementation, and which we extend to the medical domain in this thesis.

2.3 Neighbouring Pixel Relationships for Generalizable Deepfake Detection

While Frank et al. established that frequency-domain DCT representations expose GAN-generated artifacts with near-perfect linear separability, a subsequent limitation was identified by Tan et al. [44]: frequency-based representations capture *global* upsampling artifacts across the whole image, but these global patterns are insufficiently consistent across the diverse generator landscape to support robust cross-source gen-

eralisation. Tan et al. propose instead to analyse the *local* spatial consequences of upsampling, yielding a source-invariant artifact representation that achieves state-of-the-art generalisation across 28 distinct generative models spanning both GANs and diffusion architectures.

2.3.1 From Global Frequency to Local Spatial Artifacts

The core observation motivating Neighbouring Pixel Relationships (NPR) is that upsampling operations in generator pipelines, whether GAN or diffusion, produce a characteristic local interdependence among neighbouring pixels that is not adequately captured by global frequency statistics. Consider the generator’s output stage: a feature map $\hat{x} \in \mathbb{R}^{(l \times W) \times (l \times H) \times C}$ is produced by upsampling a lower-resolution map $x \in \mathbb{R}^{W \times H \times C}$ with scale factor l , after which convolutional layers with translation invariance process $\hat{x}$ to produce the output image I . When nearest-neighbour upsampling is used (scale $l = 2$), the four pixels within each 2×2 grid cell of $\hat{x}$ are initialised to identical values. After the subsequent convolutions, these values diverge, but the structural correlation introduced by the shared initialisation persists as a latent pattern in the output image. Crucially, this local correlation is present regardless of the specific upsampling strategy employed and regardless of whether the generator is a GAN or a diffusion U-Net decoder, because both architectures universally employ upsampling layers to map from a compact latent representation to full-resolution output.

This is the property that frequency-based methods [16], [21] fail to exploit robustly: the *global* frequency spectrum varies substantially across generator architectures and training datasets, as observed by Jeong et al. [26], making frequency artifacts an unreliable generalisation target. Local pixel relationships, by contrast, derive from a structural property of the upsampling operation itself and are therefore consistent across architecturally diverse generators.

2.3.2 The NPR Representation

Tan et al. formalise this intuition as the Neighbouring Pixel Relationships (NPR) representation. The image I is partitioned into non-overlapping $l \times l$ grids, each grid $v_I^c = \{w_1, \dots, w_n\}$ containing $n = l \times l$ pixel values. The NPR of a grid is computed as the element-wise difference with respect to a reference pixel w_j :

$$\hat{v}_I^c = \{w_1 - w_j, w_2 - w_j, \dots, w_n - w_j\}, \quad 1 \leq j \leq n. \quad (2.1)$$

The NPR of the complete image is the union of all grid NPRs. Setting $l = 2$ and $j = 1$ (the top-left pixel as reference) yields the default configuration, motivated by the fact that most generators employ $2\times$ upsampling layers. This representation captures the *relative, local* correlation between pixels within each upsampled grid cell and is therefore largely invariant to global image content, illumination, and semantic class. The NPR is computed directly in the spatial domain, requiring no frequency transform, and is fed as input to a lightweight ResNet-50-based binary classifier with only 1.44 million parameters.

2.3.3 Experimental Results and Relevance to Medical Detection

Tan et al. evaluate NPR on a comprehensive open-world benchmark of 28 generative models across five test datasets, encompassing 17 GANs and 11 diffusion models. All detectors are trained exclusively on ProGAN-(car, cat, chair, horse) from the Foren-Synths dataset [45] and evaluated zero-shot on all other sources. Table 2.2 reproduces their summary result.

Table 2.2: Mean detection accuracy across 28 generative models (38 sub-test sets), reproduced from Tan et al. [44]. All detectors are trained on ProGAN only. NPR outperforms all baselines by at least 11.6 percentage points, demonstrating substantially stronger cross-source generalisation.

Method	Mean Acc. (28 models)
CNNDetection [45]	57.3%
Frank [21]	56.8%
Durall [16]	56.6%
LGrad [43]	80.5%
Ojha [36]	79.8%
NPR (Tan et al.)	92.2%

Three findings from this evaluation bear directly on the medical imaging context of this thesis.

Finding 1: Frequency artifacts alone are insufficient for generalisation. Frank et al.’s frequency-based detector achieves only 56.8% mean accuracy across 28 models, marginally above chance, while NPR achieves 92.2%. This confirms the hypothesis of Jeong et al. [26] that diverse generators produce sufficiently different global frequency patterns that a detector trained to exploit one generator’s spectral fingerprint will not reliably transfer. This generalisation failure is structurally analogous to the domain gap we identify in medical imaging: the spectral gap $|\Delta\alpha|$ between real and synthetic

medical images is $4.5\times$ smaller than in natural photographs, leaving frequency-based detectors with almost no discriminative signal to exploit.

Finding 2: Local spatial artifacts extend to diffusion models. Despite being trained only on GAN-generated images, NPR achieves 95.3% mean accuracy on the DiffusionForensics benchmark [46] across eight diffusion architectures. This cross-paradigm transfer demonstrates that upsampling-induced local pixel correlations are a structural invariant of the generation process, present regardless of whether images are produced by a GAN generator or a diffusion U-Net decoder. For medical AI detection, this cross-paradigm robustness is essential: our dataset includes images from DALL-E 3, Midjourney v6, and Gemini, all of which employ diffusion-based synthesis.

Finding 3: Robustness to upsampling strategy. Tan et al. report that NPR generalises well across models employing different upsampling strategies (nearest-neighbour, bilinear, and transposed convolution), attributing this to two properties: (i) NPR is extracted from the image domain rather than from the feature map directly, and (ii) the relative, local nature of the representation is insensitive to the precise magnitude of inter-pixel differences, only their sign and relative ordering. This is a meaningful practical advantage over spectral methods, which are sensitive to the degree of anti-aliasing applied during upsampling, as demonstrated by Frank et al.’s own ablation showing that binomial upsampling reduces DCT-based detection accuracy.

NPR’s spatial-domain focus also has a direct limitation relevant to our work: it is designed to capture artifacts in the pixel neighbourhood, and does not encode the modality-level spectral context that distinguishes, for instance, a real CT scan from a real natural photograph. This is precisely the contextual signal that SPAI’s Spectral Context Vector (SCV) encodes, and that our physics-informed masking adaptation extends to acquisition-physics noise suppression. NPR and spectral learning are therefore complementary rather than competing approaches: NPR identifies where local pixel structure departs from the upsampling template; spectral learning identifies where the global frequency distribution departs from the real-image manifold.

2.4 Zero-Shot Detection via Lossless Coding Entropy

A fundamentally different paradigm for AI-generated image detection is proposed by Cozzolino et al. [11]: rather than learning to recognise the artifacts of specific generators, build a model of the real image distribution and classify any image that deviates from that model as synthetic. This *zero-shot* approach, termed ZED (Zero-shot

Entropy-based Detector), requires no synthetic images at training time and makes no assumptions about generator architectures, offering the strongest possible generalisation guarantees by construction. It is also the conceptual precursor to the paradigm adopted by SPAI, and understanding its design choices and failure modes is directly relevant to our medical domain adaptation.

2.4.1 Motivation and Conceptual Basis

The foundational insight of ZED is imported from the machine-generated text detection literature [23], [35]. Large language models generate text by sequentially sampling from learned conditional distributions over tokens. A generated sentence is therefore, by construction, *in-distribution* for its generator’s model: each token is likely given its context. A detector can exploit this by measuring the negative log-likelihood (NLL) of a test sentence under the LLM’s distribution, a low NLL signals that the sentence was likely generated by that model.

ZED translates this principle to images by replacing the LLM with a lossless image compressor. The key observation is that a well-trained lossless encoder for real images implicitly learns the probability distribution of real image pixels given their context. For a real image, the encoder’s prediction model should be accurate: the actual coding cost (NLL) should be close to the expected coding cost (entropy H). For a synthetic image, whose pixel statistics deviate from the real-image distribution, the model will be surprised: $\text{NLL} > H$ on average, or occasionally $\text{NLL} < H$ when the synthetic image is unexpectedly smooth (as observed for GLIDE images). The decision statistic is therefore the *coding cost gap* $D^{(l)} = \text{NLL}^{(l)} - H^{(l)}$, computed at each resolution level l of a multi-resolution lossless encoder.

Specifically, ZED uses the Super-Resolution based lossless Compressor (SReC) [5], which predicts each 2×2 pixel group at full resolution from a lower-resolution context via a mixture of discrete logistic distributions, enabling parallel processing and computational efficiency. At each of three resolution levels $l \in \{0, 1, 2\}$, ZED computes the pixel-wise NLL and entropy maps, spatially averages them, and forms the gap statistics $D^{(0)}$, $\Delta_{01} = D^{(0)} - D^{(1)}$, and their absolute values.

2.4.2 Experimental Results

ZED is evaluated against ten reference methods, spanning CNN-based, CLIP-based, and training-free approaches, across 30 test datasets covering 19 generative models including GANs (GauGAN, BigGAN, StyleGAN2), diffusion models (DDPM, ADM,

GLIDE, DiT, Stable Diffusion 1.4/2, SDXL), and commercial systems (DALL·E, DALL·E 2, DALL·E 3, Midjourney V5, Adobe Firefly). Using the decision statistic $|\Delta_{01}|$, ZED achieves 90.0% average AUC across all 30 datasets, representing a 3.4 percentage point improvement over the prior best competitor (Corvi2023 at 89.4% average AUC among supervised methods, Ojha2023 at 88.4%). Critically, ZED achieves this without any exposure to synthetic images at training time, while Corvi2023 is trained on 180,000 latent diffusion images.

A key advantage of ZED is threshold stability. As shown in the balanced accuracy versus threshold curves, ZED is the only method for which a single threshold provides near-optimal accuracy across all real image datasets (LSUN, FFHQ, ImageNet, COCO, RAISE, LAION). All supervised competitors exhibit multiple conflicting peaks, meaning their optimal threshold is dataset-dependent and cannot be set reliably for in-the-wild deployment.

2.4.3 Limitations and Connection to Medical Detection

Despite its strong performance on natural images, ZED’s zero-shot paradigm faces the same fundamental obstacle in the medical domain that we have identified for SPAI: its real-image model is calibrated to the statistical distribution of natural photographs.

The SReC encoder used by ZED is trained on Open Images, LAION, or COCO, all natural photograph datasets with spectral slope $\tilde{\alpha} \approx 2.5$. Real medical images, by contrast, have $\tilde{\alpha} \approx 2.87$ (chest X-rays), approaching the spectral profile of AI-generated natural images. Consequently, a ZED encoder trained on natural images would classify real medical images as *out-of-distribution* relative to its model, producing systematic false positives on clinical data. Furthermore, the coding cost gap $D^{(l)}$ measures deviation from the real-image pixel distribution at multiple resolutions. In medical imaging, acquisition-physics noise, Poisson photon counts in CT, k-space aliasing in MRI, introduces structured high-frequency content that would register as a distribution mismatch under a natural-image encoder, further inflating false positive rates on genuine clinical scans.

These limitations mirror the backbone mismatch and fixed-mask problems we identify in SPAI. Retraining ZED’s lossless encoder on sufficient medical imaging data would require a scale of curated clinical data (millions of scans) that does not currently exist in public repositories. The approach we take in this thesis, adapting the SPAI framework with a data-efficient CvT-13 backbone and modality-conditioned physics-informed masking, addresses the same root cause more tractably: by reducing the

data requirements for spectral pretraining (via CvT’s spatial inductive bias) and by explicitly suppressing acquisition-noise frequency bands before spectral comparison (via physics-informed masking).

ZED and SPAI share the same underlying theoretical motivation: real images occupy a stable, learnable region of the spectral distribution; synthetic images do not. ZED operationalises this through lossless coding entropy; SPAI through masked frequency reconstruction similarity. Both approaches require that the real-image model be calibrated to the target domain, which is precisely the adaptation challenge this thesis addresses for medical imaging.

2.5 Masked Frequency Modeling for Self-Supervised Pre-Training

The MFM framework [48] (ICLR 2023) is the foundational self-supervised pre-training method underlying the SPAI backbone, and its design choices directly shape both the strengths and the domain limitations we identify in this thesis. Understanding MFM at a mechanistic level is therefore essential to motivating our CvT-13 adaptation and physics-informed masking proposals.

2.5.1 Motivation: Spatial Redundancy in Masked Image Modeling

Masked Image Modeling (MIM) methods such as BEiT [3], MAE [24], and SimMIM [49] follow the corrupt-and-predict paradigm of BERT [13] applied to images: random image patches are masked in the spatial domain and the encoder learns to predict the missing content. This strategy is directly inspired by Masked Language Modeling (MLM), where masking out words forces the model to learn semantic context.

Xie et al. [48] identify a fundamental mismatch between language and images that limits this analogy. Unlike human-generated text, which is inherently succinct and semantically dense, raw pixel values in the spatial domain exhibit heavy spatial redundancy. A masked patch can frequently be reconstructed from its immediate spatial neighbours without the model learning any meaningful global structure. MAE compensates for this by masking an extremely high proportion of patches (75%), effectively forcing the model to reason over long spatial distances. However, this approach is architecturally constrained: operating only on visible tokens breaks the regular 2D structure of images, making MAE inapplicable to CNN architectures.

MFM addresses the redundancy problem from a different perspective: shifting the masking operation from the spatial domain to the frequency domain, where the information content is distributed more uniformly and where the characteristics of natural images are more directly observable.

2.5.2 The MFM Pre-Training Pipeline

The MFM pipeline consists of four components, illustrated in Figure 2.3.

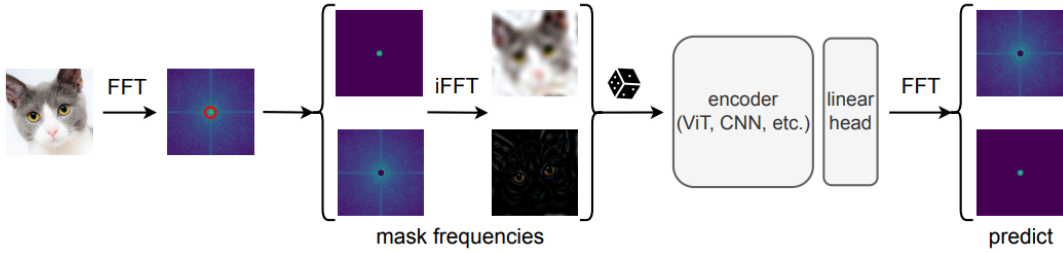


Figure 2.3: Overview of the MFM pre-training pipeline (from Xie et al. [48]). Each input image is converted to the frequency domain via 2D FFT. A low-pass or high-pass circular mask is applied to the frequency spectrum, and the inverse FFT produces a spatially corrupted image that is fed to the encoder. A lightweight linear decoder then predicts the masked frequency values on the spectrum using a frequency reconstruction loss. Because no mask tokens are inserted, the same pipeline applies to both ViT and CNN encoders without architectural modification.

Masking strategy. Given an input image $x \in \mathbb{R}^{H \times W}$, the 2D Discrete Fourier Transform $\mathcal{F}(x)$ maps it to a complex-valued frequency spectrum. A circular binary mask $M \in \{0, 1\}^{H \times W}$ is defined by a radius hyperparameter r : positions within Euclidean distance r of the spectrum centre are set to 1 (low-frequency band); all others to 0. Low-pass filtering retains $M \odot \mathcal{F}(x)$ and discards the high-frequency content; high-pass filtering retains $(1 - M) \odot \mathcal{F}(x)$. At each training step, the filter type is sampled from a Bernoulli distribution with equal probability, so the model must learn to reconstruct both low- and high-frequency components. The filtered spectrum is converted back to the spatial domain via inverse FFT, yielding a corrupted image with either fine structure removed (low-pass) or colour and smooth gradients removed (high-pass).

Encoder and decoder. Because no mask tokens are inserted, the encoder receives a spatially corrupted but structurally complete image, a critical property that makes MFM architecture-agnostic. Standard ViT patch embeddings and CNN sliding-window convolutions both operate normally on the corrupted input. The decoder is a single linear layer, avoiding the overhead of a deep transformer decoder while achieving competitive performance.

Reconstruction target and loss. The decoder predicts the masked frequency values on the spectrum. The loss is a frequency distance metric defined over the real and imaginary components of the complex spectrum:

$$\mathcal{L} = \frac{1}{HW} \sum_{u,v} |\mathcal{F}_r(x)(u, v) - \mathcal{F}_o(x)(u, v)|^\gamma, \quad (2.2)$$

where $\mathcal{F}_r$ and $\mathcal{F}_o$ are the reconstructed and original spectra, $\gamma = 1$ by default, and the sum is restricted to the masked region. Predicting only the masked area outperforms full-spectrum reconstruction, consistent with the MIM principle that predicting invisible signals is a more effective self-supervisory task.

2.5.3 Key Empirical Findings

Ablations on ImageNet-1K with ViT-B/16 (300-epoch pre-training, top-1 fine-tuning accuracy) yield three findings directly relevant to our work.

Both frequency bands carry useful information. Masking high frequencies only (low-pass input) achieves 82.4%; masking low frequencies only (high-pass input) achieves 82.3%; random sampling of both achieves 83.1%. This confirms that high-frequency components, which encode object outlines and fine structural detail, and low-frequency components, which encode smooth gradients, colour, and global layout, are complementary sources of representational signal. This finding is directly consequential for medical imaging: the acquisition physics of CT, MRI, and X-ray modulate the high-frequency band via Poisson photon noise and k-space aliasing, inserting modality-specific structured content that a fixed circular mask at a single radius cannot distinguish from synthesis artifacts.

A frequency loss outperforms a spatial loss. Replacing the frequency distance of Equation 2.2 with a spatial ℓ_1 or ℓ_2 loss reduces top-1 accuracy by 0.8–0.9 percentage points. This demonstrates that alignment between the reconstruction objective and the frequency domain representation is critical, a principle we extend in our physics-informed masking, which conditions the mask on modality-specific frequency structure rather than applying a fixed geometric partition.

Architecture-agnostic pre-training via frequency masking. Applying spatial random-patch masking (MIM) to ResNet-50 yields 77.7% top-1 accuracy, below the supervised RSB A3 baseline of 78.1%, because mask tokens are incompatible with CNN’s dense sliding-window paradigm and information leaks across convolutional layers [17]. MFM raises ResNet-50 to 78.5%, surpassing the supervised baseline. This

architecture-agnosticism is the primary reason SPAI adopts the MFM-pretrained ViT-B/16 as its frozen backbone: the same pre-training recipe could in principle support alternative encoders. Our CvT-13 adaptation exploits precisely this property, replacing the ViT backbone with a data-efficient hybrid CNN-Transformer while retaining the MFM pre-training paradigm.

2.5.4 Relation to Low-Level Image Restoration

Xie et al. conduct the first systematic study of low-level image processing tasks, super-resolution (SR), deblurring, and denoising, as self-supervised pre-training objectives, analysed from a unified frequency perspective. SR and deblurring remove predominantly high-frequency content via downsampling and Gaussian blur; denoising corrupts the full frequency spectrum. Under the same training protocol as MFM, denoising (82.7%) outperforms SR (82.4%) and deblurring (81.7%), because it activates the full spectrum and thus brings it closer to MFM’s random low-/high-pass sampling strategy.

Critically, SR and deblurring produce performance *degradation* on ResNet-50 relative to the supervised baseline, while MFM consistently improves both architectures. The authors attribute this to the inductive bias of convolution: CNNs are already effective at capturing high-frequency components via their sliding-window kernels, so pretraining that focuses on high-frequency reconstruction provides no complementary signal. This architectural asymmetry is directly relevant to our CvT-13 adaptation: CvT introduces local convolutional projections within each transformer stage, partially restoring the CNN inductive bias for high-frequency structure. Our choice of CvT must therefore be paired with careful attention to the masking strategy, ensuring that both frequency bands remain informative pre-training targets despite the encoder’s increased high-frequency sensitivity.

2.5.5 Connection to SPAI and the Medical Domain Limitation

SPAI [27] inherits the MFM pre-training protocol directly: a ViT-B/16 is pre-trained via masked frequency reconstruction on ImageNet-1K (1.2 million natural photographs) and then frozen during downstream detection training. The frozen backbone is used to extract representations at each of 12 transformer blocks, and the cross-band consistency of these representations (Spectral Reconstruction Similarity, SRS) serves as the primary detection signal.

This design choice introduces two domain limitations for medical imaging that our

thesis addresses. First, the spectral prior encoded in the frozen backbone reflects the $1/f^\alpha$ statistics of natural photographs ($\bar{\alpha} \approx 2.5$), not the higher spectral slopes of clinical modalities ($\bar{\alpha} \approx 2.87$ for chest X-rays). A backbone whose spectral prior is miscalibrated to the target domain will produce SRS signals that are unreliable indicators of synthesis. Second, the fixed circular mask at radius $r = 16$, covering only 1.6% of all frequency bins, partitions the spectrum without regard to modality-specific acquisition noise bands, conflating Poisson photon statistics in CT and k-space aliasing in MRI with the absence of high-frequency content characteristic of AI-generated images. Our physics-informed frequency masking addresses the second limitation by learning a modality-conditioned mask that suppresses known acquisition-noise bands before the spectral comparison is computed.

2.6 SPAI: Any-Resolution AI-Generated Image Detection by Spectral Learning

Karageorgiou et al. [27] (CVPR 2025) propose *Spectral AI-generated Image detection* (SPAI), the direct foundation of this thesis. SPAI departs from the dominant paradigm of learning to recognise generator-specific artifacts and instead builds a spectral model of the real image distribution, detecting AI-generated images as out-of-distribution samples of that model. This section presents the architecture in full technical detail, as every component we adapt in Chapters 3 and 4 is introduced here.

2.6.1 Core Principle: Invariance of the Real Spectral Distribution

A fundamental observation motivates SPAI: while the artifacts introduced by different generative models vary significantly [2], the spectral distribution of real images is not directly related to the introduction of any specific generative model. Instead, it is determined by long-term technological factors, optics, sensors, and image processing pipelines, constituting an invariant for the detection task. AI-generated images, regardless of the generator used, constitute out-of-distribution samples with respect to this invariant spectral model. This reframing sidesteps the fundamental generalisation problem of artifact-based detectors: because the real spectral distribution does not shift when new generators are introduced, a detector calibrated to that distribution remains valid across the open-ended generator landscape.

An overview of the full SPAI architecture is shown in Figure 2.4.

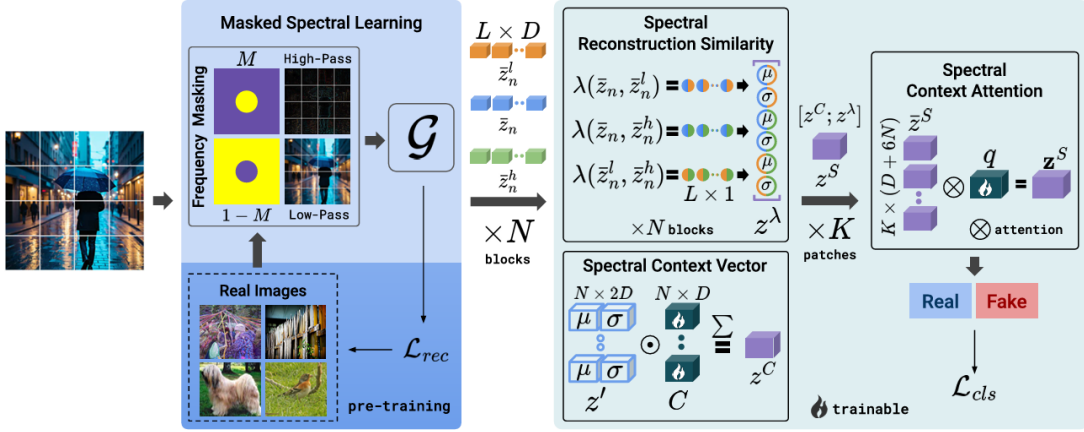


Figure 2.4: Overview of the SPAI architecture (from Karageorgiou et al. [27]). A MFM-pretrained ViT-B/16 backbone (kept frozen) encodes the spectral distribution of real images. Spectral Reconstruction Similarity (SRS) measures the divergence between representations of an image and its frequency-filtered components at each transformer block. The Spectral Context Vector (SCV) captures the spectral context under which SRS values are computed. Spectral Context Attention (SCA) fuses patch-level spectral vectors into an image-level representation with $\mathcal{O}(K)$ complexity, enabling any-resolution processing. A three-layer MLP produces the final binary prediction.

2.6.2 Masked Spectral Learning

To build a spectral model of real images, SPAI employs the frequency reconstruction pretext task introduced by MFM [48] under a self-supervised setup using only real images.

Frequency masking. Given an input image $x \in \mathbb{R}^{H \times W}$, the 2D Discrete Fourier Transform yields a complex-valued spectrum $\chi = \mathcal{F}(x) \in \mathbb{C}^{H \times W}$, where $\chi(u, v)$ is the complex frequency value at coordinates (u, v) . A binary mask $M \in \{0, 1\}^{H \times W}$ is defined by a fixed radius r from the spectrum centre (c_H, c_W) :

$$M(u, v) = \begin{cases} 0 & \text{if } d((u, v), (c_H, c_W)) < r \\ 1 & \text{otherwise,} \end{cases} \quad (2.3)$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance. The high- and low-frequency components of the image are then:

$$x^h = \mathcal{F}^{-1}(\chi \odot M), \quad (2.4)$$

$$x^l = \mathcal{F}^{-1}(\chi \odot (\mathbf{1} - M)), \quad (2.5)$$

where $\mathbf{1}$ is the all-ones matrix and $\odot$ denotes point-wise multiplication.

An auxiliary decoding head $\mathcal{H}$ is trained alongside the backbone $\mathcal{G}$ to reconstruct the input from one of the two filtered versions, selected randomly by a Bernoulli distribution $B(\cdot, \cdot)$:

$$\hat{x} = \mathcal{H}(\mathcal{G}(B(x^l, x^h))) . \quad (2.6)$$

The reconstruction loss is the frequency distance $\mathcal{L}_{\text{rec}} = D(\mathcal{F}(x), \mathcal{F}(\hat{x}))$ introduced by MFM [48], and $\mathcal{H}$ is discarded after pre-training. SPAI uses a ViT-B/16 [14] pre-trained via this procedure on ImageNet-1K (1.2 million images) with masking radius $r = 16$, and keeps its weights *frozen* throughout all downstream training. The choice of backbone is validated in Table 2.3, reproduced from the original paper.

Table 2.3: Evaluation of different pre-training backbones within SPAI, reproduced from Karageorgiou et al. [27]. Average AUC over 5 sources of real images and 13 generative models is reported. The MFM-pretrained ViT-B/16 outperforms both CLIP and DINOv2 despite being pre-trained on $117\times$ and $333\times$ less data respectively.

Backbone	Pre-training Data	AUC (%)
CLIP ViT-B/16	400 million images	87.6
DINOv2 ViT-B/14	142 million images	87.5
MFM ViT-B/16 (SPAI)	1.2 million images	91.0

2.6.3 Spectral Reconstruction Similarity

Once $\mathcal{G}$ has been pre-trained, its block-wise representations serve as the detection signal. Let $\mathcal{G}$ comprise N transformer blocks. Following the ViT tokenization procedure, input x is split into $L = HW/p^2$ tokens of size $p \times p$, each linearly projected to a d -dimensional embedding, forming the input sequence $z_0 \in \mathbb{R}^{L \times d}$. The output of the n -th transformer block is:

$$z_n = \mathcal{G}_n(z_0) \in \mathbb{R}^{L \times d}, \quad n \in \{1, \dots, N\}. \quad (2.7)$$

The original image x , its high-pass component x^h , and its low-pass component x^l are each encoded via Equation 2.7, yielding z_n , z_n^h , and z_n^l respectively.

N projection operators $\mathcal{P}_n : \mathbb{R}^d \rightarrow \mathbb{R}^D$, one per transformer block, then project these representations into a shared comparison space. Each operator applies LayerNorm, followed by a linear layer, GELU activation, and a second linear layer with LayerNorm. The projected representations are denoted $\bar{z}_n$, $\bar{z}_n^h$, and $\bar{z}_n^l \in \mathbb{R}^{L \times D}$.

SRS definition. For two representations $z^A, z^B \in \mathbb{R}^{L \times D}$, the Spectral Reconstruction

Similarity is defined as the token-wise cosine similarity:

$$\lambda(z^A, z^B) = \frac{z^A \cdot z^B}{\|z^A\| \|z^B\|} \in [-1, 1]^{L \times 1}. \quad (2.8)$$

Three SRS vectors are computed at each block n :

$$\omega_n^{ol} = \lambda(\bar{z}_n, \bar{z}_n^l), \quad (2.9)$$

$$\omega_n^{oh} = \lambda(\bar{z}_n, \bar{z}_n^h), \quad (2.10)$$

$$\omega_n^{lh} = \lambda(\bar{z}_n^l, \bar{z}_n^h). \quad (2.11)$$

Each SRS vector has L entries, one per ViT token. Their mean and standard deviation are computed, yielding two scalars per SRS vector per block. Concatenating these six values across all N blocks gives the SRS feature vector:

$$z^\lambda \in [-1, 1]^{6N}. \quad (2.12)$$

The intuition is direct: since $\mathcal{G}$ has learned the spectral distribution of real images, its block-wise representations of x , x^l , and x^h will be mutually consistent for real images (high λ values) but inconsistent for AI-generated images (low λ values), whose spectral distribution is out-of-distribution for the frozen model.

2.6.4 Spectral Context Vector

Not all SRS values are equally informative for all images. On images with little high-frequency content, SRS values relating to high-frequency reconstruction carry almost no signal. To capture the spectral context under which z^λ is computed, SPAI introduces the *Spectral Context Vector* (SCV) $z^C \in \mathbb{R}^D$.

Starting from the projected representations $\bar{z}_n$ (Sec. 2.6.3), the mean and standard deviation are computed over the L ViT tokens, producing two D -dimensional vectors per block. These are concatenated across all N blocks into $z' \in \mathbb{R}^{N \times 2D}$.

A learnable spectral map $C \in \mathbb{R}^{N \times D}$ and two projection functions $\mathcal{P}_1 : \mathbb{R}^{2D} \rightarrow \mathbb{R}^D$ and $\mathcal{P}_2 : \mathbb{R}^D \rightarrow \mathbb{R}^D$ then construct the SCV via a soft attention over the per-block feature summaries:

$$C' = \mathcal{P}_2(\text{softmax}(C) \odot \mathcal{P}_1(z')) \in \mathbb{R}^{N \times D}, \quad (2.13)$$

$$z^C = \sum_{n=1}^N C'_n \in \mathbb{R}^D. \quad (2.14)$$

The SCV and SRS vectors are concatenated into the *spectral vector*:

$$z^S = [z^C; z^\lambda] \in \mathbb{R}^{D+6N}. \quad (2.15)$$

2.6.5 Spectral Context Attention

Modern photographs may be many megapixels in size. ViT’s quadratic self-attention complexity prohibits direct processing at native resolution: on an Nvidia L40S 48 GB GPU, standard self-attention is limited to approximately 18 megapixels before running out of memory. SPAI addresses this through *Spectral Context Attention* (SCA), which processes images of arbitrary resolution in $\mathcal{O}(K)$ time.

The input image is split into K non-overlapping patches of size $h \times w$ pixels, forming the patched representation $x^P \in \mathbb{R}^{K \times h \times w}$. For each patch x_k , a spectral vector z_k^S is computed following Equations 2.8–2.15. These are stacked into $\bar{z}^S \in \mathbb{R}^{K \times (D+6N)}$.

Defining weight matrices $W_K, W_V \in \mathbb{R}^{(D+6N) \times D_h}$ and $W_O \in \mathbb{R}^{D_h \times (D+6N)}$, and a learnable spectral reconstruction importance vector $q \in \mathbb{R}^{D_h}$, the patch-level vectors are fused into a single image-level spectral vector via cross-attention:

$$\mathcal{A} = \text{softmax} \left(\frac{q \cdot (\bar{z}^S W_K)^\top}{\sqrt{D_h}} \right) \in (0, 1)^{1 \times K}, \quad (2.16)$$

$$z^S = (\mathcal{A} \cdot (\bar{z}^S W_V)) W_O \in \mathbb{R}^{D+6N}. \quad (2.17)$$

Because q is a single vector, the key computation collapses to a $1 \times K$ attention map, linear in the number of patches. During training, $K_{\text{train}} = 4$ augmented random views of the input image serve as patches, allowing any image to participate in training regardless of resolution. During inference, the actual K patches at the native resolution are used.

The classification head is a three-layer MLP with ReLU activations on the first two layers and a sigmoid on the final one, operating on z^S to produce the binary prediction $\hat{y} \in (0, 1)$. Training minimises binary cross-entropy:

$$\mathcal{L}_{\text{cls}} = \text{BCE}(\hat{y}, y). \quad (2.18)$$

2.6.6 State-of-the-Art Performance and Robustness

SPAI is trained on 180k low-resolution images from a single latent diffusion model [9] paired with 180k real images, and evaluated zero-shot on 13 unseen generators spanning GAN, diffusion, and commercial architectures at resolutions from 0.1 to 4.1 megapixels. Table 2.4 reproduces the main comparison from the paper.

Table 2.4: Comparison of AI-generated image detectors across 13 generative models, reproduced from Karageorgiou et al. [27]. Average AUC over 5 sources of real images is reported. SPAI achieves the highest average AUC across all image sizes, with a 5.5% absolute improvement over the prior state-of-the-art RINE [30].

Method	<0.5 MP			0.5–1.0 MP					>1.0 MP					AVG
	Glide	SD1.3	SD1.4	Flux	DALLE2	SD2	SDXL	SD3	GigaGAN	MJv5	MJv6.1	DALLE3	Firefly	
NPR [44]	72.2	89.6	60.5	19.8	3.9	12.5	18.1	60.6	83.2	15.3	19.8	97.1	38.0	45.4
CNNDet. [45]	59.2	59.0	61.2	39.8	71.5	57.5	67.4	30.2	73.4	48.8	56.7	23.5	73.4	55.5
LGrad [43]	76.5	82.4	83.4	74.9	85.7	60.7	70.2	12.7	89.9	69.2	79.6	30.0	42.0	65.9
DMID [9]	73.1	100.0	100.0	97.2	54.3	99.7	99.6	67.9	67.9	99.9	94.4	41.3	90.2	83.5
RINE [30]	95.6	99.9	99.9	93.0	93.0	96.6	99.3	39.1	92.9	96.4	81.2	41.8	82.9	85.5
SPAI	90.2	99.6	99.6	83.0	91.1	96.5	97.4	75.9	85.4	94.5	84.0	90.2	96.0	91.0

2.6.7 Why SPAI Fails on Medical Images

Despite this strong performance on natural images, three structural properties of SPAI directly cause its failure in the medical imaging domain, as quantified in Section 1.3.

Backbone spectral miscalibration. The frozen MFM-pretrained ViT-B/16 encodes a spectral prior calibrated to the $1/f^\alpha$ statistics of natural photographs ($\bar{\alpha} \approx 2.5$). Real chest X-rays have $\bar{\alpha} \approx 2.87$, approaching the spectral profile of AI-generated natural images. The SRS signals produced by a backbone calibrated to $\alpha \approx 2.5$ are unreliable indicators of synthesis for data drawn from a population with $\alpha \approx 2.87$.

Fixed circular mask. The mask defined in Equation 2.3 at radius $r = 16$ covers only approximately 1.6% of all frequency bins and applies a single fixed geometric partition regardless of imaging modality. Acquisition-physics noise in CT (Poisson photon statistics) and MRI (k-space aliasing) produces structured high-frequency content that is spectrally indistinguishable from the absence of high-frequency content in AI-generated images under this fixed partition, producing systematic false negatives.

Narrow spectral discrimination gap. The $|\Delta\alpha|$ detection signal, the difference in spectral slope between real and synthetic images, is $4.5\times$ smaller in the medical domain ($|\Delta\alpha| = 0.170$) than in natural photographs ($|\Delta\alpha| = 0.775$). The SRS values computed by Equations 2.9–2.11 depend on this gap: when real and synthetic images are spectrally nearly identical, the cosine similarities ω_n^{ol} , ω_n^{oh} , and ω_n^{lh} converge for both classes, collapsing the decision boundary.

These three failure modes motivate the three adaptations proposed in Chapter 3: CvT-13 backbone substitution, physics-informed frequency masking, and the novel medical AI-generated image dataset.

2.7 Convolutional Vision Transformer (CvT)

Wu et al. [47] propose the *Convolutional vision Transformer* (CvT), a hybrid architecture that introduces convolutions into the Vision Transformer (ViT) at two structural levels, yielding a model that inherits the local inductive bias of CNNs while retaining the global context and dynamic attention of Transformers. CvT is the backbone we substitute into the SPAI framework in Chapter 3, and understanding its architectural properties is essential to motivating that choice.

2.7.1 Motivation: Limitations of Pure Vision Transformers

ViT and its derivatives [14] process images by splitting them into non-overlapping 16×16 patches, linearly projecting each patch into a fixed-dimensional token, and modelling all inter-token relationships via global self-attention. This design achieves excellent performance at large scale but exhibits a well-known data efficiency weakness: because all spatial relationships must be learned from scratch via attention, ViT requires very large datasets (ImageNet-22k or JFT-300M) to converge to competitive accuracy. On smaller datasets, CNN counterparts such as ResNet outperform ViT comfortably, because convolutional kernels encode two structural priors that reduce the number of examples needed to learn useful representations: *local receptive fields* (neighbouring pixels are correlated and should be processed jointly) and *weight sharing* (the same filter is applied at all spatial positions, leveraging translation equivariance).

For our medical imaging context this data-efficiency gap is critical. As established in Section 1.3, retraining SPAI’s MFM-pretrained ViT-B/16 backbone from scratch on medical data would require 1.2–14 million images, a scale unavailable in public clinical repositories. A backbone with built-in spatial inductive bias can be pretrained on substantially fewer images while retaining competitive spectral representations, precisely the property CvT is designed to provide.

2.7.2 Architectural Design

CvT introduces convolutions at two points in the Transformer pipeline, illustrated in Figure 2.5.

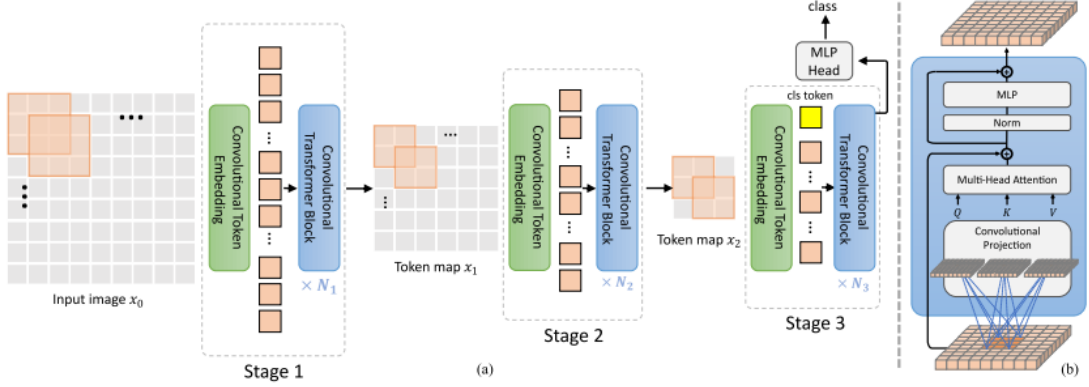


Figure 2.5: Overview of the CvT architecture (from Wu et al. [47]). **(a)** The hierarchical multi-stage pipeline: each stage begins with a Convolutional Token Embedding layer that performs an overlapping strided convolution on the 2D-reshaped token map, progressively reducing spatial resolution while increasing feature dimension. **(b)** The Convolutional Transformer Block: the standard position-wise linear projection for queries, keys, and values is replaced with a depth-wise separable Convolutional Projection, enabling local spatial context modelling within the attention mechanism. The classification token is appended only in the final stage.

Convolutional Token Embedding. At the beginning of each stage i , the 2D-reshaped token map $x_{i-1} \in \mathbb{R}^{H_{i-1} \times W_{i-1} \times C_{i-1}}$ is processed by a learned convolution $f(\cdot)$ with kernel size $s \times s$, stride $s - o$, and padding p , producing a new token map $f(x_{i-1}) \in \mathbb{R}^{H_i \times W_i \times C_i}$, where:

$$H_i = \left\lfloor \frac{H_{i-1} + 2p - s}{s - o} \right\rfloor + 1, \quad W_i = \left\lfloor \frac{W_{i-1} + 2p - s}{s - o} \right\rfloor + 1. \quad (2.19)$$

This overlapping convolution simultaneously reduces the token sequence length and increases the feature dimension across stages, mirroring the spatial downsampling and channel expansion of CNN feature pyramids. Crucially, the overlapping receptive field captures local spatial structure that non-overlapping patch embeddings (as in ViT) discard entirely.

Convolutional Projection. Within each Transformer block, the standard position-wise linear projection that produces query, key, and value matrices is replaced with a depth-wise separable convolution of kernel size $s \times s$:

$$x_i^{q/k/v} = \text{Flatten}(\text{Conv2d}(\text{Reshape}_{2D}(x_i), s)), \quad (2.20)$$

where Conv2d denotes depth-wise separable convolution implemented as Depth-wise

Conv2d $\rightarrow$ BatchNorm2d $\rightarrow$ Point-wise Conv2d. For key and value projections, a stride of 2 is used by default (the *squeezed* configuration), reducing the key and value token count by 4 $\times$ and thereby reducing the Multi-Head Self-Attention FLOPs by 4 $\times$, with only a marginal 0.3% drop in ImageNet top-1 accuracy.

No positional embedding. Because both the Convolutional Token Embedding and the Convolutional Projection encode spatial relationships implicitly through their local receptive fields and translation-equivariant weight sharing, CvT does not require an explicit positional embedding. Wu et al. demonstrate that removing positional embeddings from CvT does not degrade ImageNet accuracy (Table 5 in the original paper), whereas removing them from DeiT causes a 1.8% accuracy drop. This property is directly relevant to our application: the absence of fixed positional embeddings means CvT operates correctly on inputs of variable resolution without interpolation, making it naturally resolution-agnostic, a property SPAI’s SCA component requires of its backbone.

2.7.3 Model Variants and Efficiency

Wu et al. instantiate three CvT variants by varying the number of Transformer blocks per stage and the per-stage feature dimension. CvT-13 (19.98 M parameters, 4.53 G FLOPs) and CvT-21 (31.54 M parameters, 7.13 G FLOPs) are the basic models; CvT-W24 (276.7 M parameters) is a wide variant for large-scale pretraining. On ImageNet-1k, CvT-13 achieves 81.6% top-1 accuracy, outperforming PVT-Small, T2T-ViT_t-14, and TNT-S while using fewer parameters, and surpassing ResNet-152 by 3.2% at one-third the parameter count.

We adopt **CvT-13** as our backbone substitution for two reasons. First, its 19.98 M parameters represent a 4.3 $\times$ reduction relative to SPAI’s ViT-B/16 backbone (85.8 M), directly addressing the parameter efficiency requirement for data-scarce medical pre-training. Second, at 4.53 G FLOPs on 224 $\times$ 224 inputs, CvT-13 is computationally comparable to ViT-B/16’s processing footprint per patch, preserving compatibility with SPAI’s patch-level SCA inference pipeline.

2.7.4 Relevance to SPAI Backbone Substitution

Three properties of CvT make it a well-motivated replacement for SPAI’s frozen MFM-pretrained ViT-B/16 backbone in the medical domain.

Spatial inductive bias reduces data requirements. The convolutional projections encode local pixel correlations by construction, reducing the number of training ex-

amples the model must observe before learning useful spatial representations. This is the key property that enables MFM pretraining on ImageNet-Mini (a $21\times$ smaller dataset than the full ImageNet-1K used by SPAI’s ViT-B/16) while retaining competitive spectral representations, as validated in our pilot experiments (Section 4.3).

Hierarchical multi-scale processing. CvT’s three-stage pyramid produces feature maps at 56×56 , 28×28 , and 14×14 spatial resolutions, in contrast to ViT-B/16’s single-scale 14×14 token grid. These multi-scale representations are aggregated by our SpatialChannelAdapter module (described in Chapter 3) to produce the block-wise representations consumed by SPAI’s SRS computation.

Resolution-agnostic operation. The absence of positional embeddings in CvT means the model accepts arbitrary spatial input sizes without interpolation artifacts, preserving compatibility with SPAI’s any-resolution SCA inference pipeline, which processes high-resolution images as a set of 224×224 patches.

2.8 AI-Generated Medical Image Detection: Domain-Specific Prior Work

The preceding sections establish the spectral and architectural foundations for AI-generated image detection on natural photographs. This section surveys work that addresses the medical domain directly, tracing the threat landscape from early adversarial manipulation attacks through to the most recent multi-modal forensic detectors. Taken together, these works confirm that the natural-image paradigm does not transfer to clinical imagery, and that domain-specific datasets, architectures, and evaluation protocols are essential, the precise gaps our thesis addresses.

2.8.1 CT-GAN: The Medical Manipulation Threat

The medical deepfake threat was made concrete by Mirsky et al. [34], who demonstrated that a 3D conditional GAN (*CT-GAN*) can inject or remove evidence of lung cancer in volumetric CT scans with millisecond latency. CT-GAN achieves this by learning two paired generators: one trained to inject cancerous nodules into healthy scans, and another trained to erase nodules from malignant ones. The attack does not require retraining for each patient; it operates by inpainting a learned nodule texture into a target region of any input volume.

The authors evaluated CT-GAN against three expert radiologists and a state-of-the-art AI lung-cancer screening system in a blinded study. Radiologists misclassified 99% of

injected scans as malignant, and 94% of nodule-removed scans as healthy. After being informed of the attack, their accuracy on tampered images remained poor (60% and 87% error rates respectively), demonstrating that visual inspection alone is insufficient for forgery detection in 3D volumetric data. To demonstrate the practical attack surface, the authors conducted a covert penetration test on an active hospital network, intercepting and modifying CT scans in transit between the scanner and the PACS server.

CT-GAN established two findings that are foundational for our work. First, the threat of medical image forgery is not theoretical: the same GAN architectures used in natural-image deepfakes can be repurposed for clinical manipulation with minimal adaptation. Second, existing clinical workflows, including radiologist review and AI diagnostic tools, offer no reliable defence, motivating the need for dedicated forensic detectors that operate at the image level. CT-GAN’s manipulation paradigm (local inpainting of pathological content) is qualitatively different from the full-image generation scenario we address in this thesis, but it establishes the severity of the threat and the inadequacy of human inspection as a countermeasure.

2.8.2 M3Dsynth: A Benchmark for CT Manipulation Detection

Building directly on the CT-GAN threat model, Zingarini et al. [52] introduced M3Dsynth, a large-scale dataset of manipulated CT lung images designed to support the development and benchmarking of forensic detectors. M3Dsynth contains 8,577 manipulated CT volumes, generated by injecting or removing lung cancer nodules using three distinct methods spanning both GAN-based and diffusion-model-based synthesis pipelines.

A key empirical finding is that the manipulated images fool state-of-the-art automated diagnostic tools, replicating the vulnerability demonstrated by Mirsky et al. but at dataset scale. The authors then evaluated several existing forensic detectors on M3Dsynth and showed that, once retrained on the dataset, these detectors can accurately detect and localise manipulated content, including in cross-generator evaluation settings where the test manipulation method differs from the training method. This cross-generator generalisation result directly parallels the natural-image findings of Wang et al. [45], but confined to the CT domain.

M3Dsynth is relevant to our work in two respects. First, it confirms that the localised manipulation scenario requires its own evaluation benchmark: the forensic signal in a locally inpainted CT slice is concentrated in the boundary between the real tissue and

the inpainted nodule, rather than in the global spectral statistics that SPAI exploits. Second, the dataset demonstrates that cross-generator generalisation is achievable in the medical domain given sufficient training signal, a result that motivates our construction of a multi-generator medical benchmark (Section 3.4) covering full-image synthesis rather than local manipulation.

2.8.3 AFFETDS: Adversarial Feature Fusion for MRI Tumour Manipulation Detection

Ali et al. [1] propose AFFETDS (Adversarial Feature Fusion Enhanced Tumour Detection System), an ensemble framework targeting the specific sub-problem of tumour insertion and removal forgery in MRI scans. AFFETDS combines three mechanisms: adversarial training using Projected Gradient Descent (PGD) and the Fast Gradient Sign Method (FGSM) to harden the model against adaptive attacks; feature fusion that concatenates deep features extracted by ResNet-50 with handcrafted Histogram of Oriented Gradients (HOG) descriptors; and a Support Vector Machine (SVM) ensemble classifier with weighted voting.

Evaluated on a dataset of 1,378 MRI scans (774 real, 604 manipulated) drawn from the TCIA and ADNI repositories, AFFETDS achieves 91.5% accuracy, 90.7% precision, and 91.2% recall, outperforming purely deep-learning baselines (SVM: 86.2%; CNN: 88.4%). The authors attribute the gain of the hybrid feature fusion over either modality alone to the complementary nature of the two representations: ResNet-50 features capture broad semantic patterns and global texture, while HOG captures the local gradient structure that is clinically meaningful in MRI tissue boundaries and is specifically disrupted by GAN-based inpainting.

Two aspects of AFFETDS are directly relevant to our work. First, the adversarial training strategy demonstrates that MRI forensics classifiers are susceptible to gradient-based evasion attacks, a vulnerability that purely spectral approaches such as SPAI also inherit, as noted by Frank et al. [21] and Carlini and Farid [6]. Second, the fusion of handcrafted frequency-sensitive features (HOG) with learned deep representations echoes our motivation for physics-informed frequency masking: both approaches argue that domain-specific signal structure in medical images must be explicitly encoded rather than left to the network to discover. AFFETDS, however, is evaluated exclusively on MRI tumour manipulation and does not address full-image synthesis, multi-modality generalisation, or the spectral domain gap we quantify in Section 1.3.

2.8.4 DSKI and MedForensics: Towards Multi-Modal Medical Forensics

The most comprehensive recent contribution to medical deepfake detection is Li et al. [31] (MICCAI 2025), which introduces both a large-scale benchmark and a novel detector designed specifically for AI-generated medical images.

MedForensics Dataset. MedForensics covers six imaging modalities (CT, X-ray, MRI, ultrasound, endoscopy, and pathology) and twelve state-of-the-art generative models, making it the first benchmark to span the breadth of clinical imaging modalities at this scale. The dataset addresses a critical gap identified by prior work: existing forensic benchmarks are either restricted to natural images, limited to a single modality, or generated by a narrow set of architectures.

DSKI Detector. The Dual-Stage Knowledge Infusing (DSKI) detector is built on a CLIP ViT-L/14 backbone and injects medical forensic knowledge through two complementary mechanisms. In the training stage, a Cross-Domain Fine-Trace Adapter (CDFA) is inserted into three transformer blocks of the frozen CLIP backbone. The CDFA extracts forensic clues from both the spatial domain (via an inception module for multi-scale artifact capture) and the noise residual domain (via a constrained CNN for low-level pixel statistics), fusing them into an enriched feature representation. In the testing stage, a Medical Forensic Retrieval Module (MFRM) augments detection accuracy through few-shot retrieval from a feature bank of real and synthetic medical images: for each test image, the n most similar examples are retrieved and their labels are aggregated with the backbone prediction. DSKI significantly outperforms existing state-of-the-art methods across all six modalities in MedForensics, achieving over 20% improvement in accuracy and average precision relative to the UniFD baseline that uses the unmodified CLIP feature space. A blinded evaluation with three domain specialists (a radiologist, a pathologist, and a gastroenterologist) further confirms that DSKI surpasses human expert performance on the detection task.

Relation to Our Work. DSKI and our proposed framework share the core motivation that natural-image forensic methods are inadequate for clinical data and that domain-specific adaptation is required. However, the two approaches differ in their adaptation strategies. DSKI injects knowledge through learned cross-domain adapters inserted into a CLIP backbone that was pretrained on 400 million image-text pairs, a substantially larger pretraining corpus than the 1.2 million images used by SPAI’s MFM-pretrained ViT-B/16, or the ImageNet-Mini corpus used in our CvT-13 pilot.

Our adaptation instead pursues data efficiency through architectural inductive bias (CvT-13’s convolutional projections) and explicit physics-informed masking that targets the spectral statistics of individual acquisition modalities. Whereas DSKI’s MFRM addresses generalisation at test time through retrieval augmentation, our approach addresses it during training through a modality-conditioned mask that suppresses acquisition-noise frequency bands specific to CT, MRI, and X-ray imagery. Furthermore, DSKI is evaluated on a multi-modality benchmark but does not report spectral analysis of the domain gap, leaving the mechanistic root cause of detection difficulty in medical images unquantified. Our spectral analysis (Section 4.2) provides precisely this quantification, establishing that the $|\Delta\alpha|$ gap is $4.5\times$ smaller in medical images, and connecting this measurement directly to the observed near-chance zero-shot accuracy of frequency-based detectors.

Chapter 3

Proposed Methodology

Our methodology adapts the SPAI detection framework to the medical imaging domain through four coordinated components, each addressing a specific identified failure mode identified in Chapter 2.

3.1 Backbone Replacement: ViT-B/16 to CvT-13

SPAI’s MFM-pretrained ViT-B/16 backbone was pretrained on ImageNet, whose spectral statistics differ substantially from medical imagery. Retraining ViT-B/16 via MFM on medical data would require 1.2–14 million medical images, a dataset scale that is not feasibly available in clinical repositories.

We propose replacing the ViT-B/16 backbone with CvT-13 (Convolutional Vision Transformer, Wu et al., 2021), a hybrid architecture combining convolutional inductive biases with Transformer global modelling. CvT-13 offers three key advantages over ViT-B/16 for the medical domain.

Data efficiency. Convolutional inductive bias (locality, translation equivariance) allows the model to learn meaningful spectral features from far fewer training examples, making it tractable to pretrain on substantially smaller medical datasets than those required by a pure Vision Transformer.

No positional encoding. Position is encoded implicitly in convolutional operations, enabling native handling of variable medical image resolutions without resizing or positional embedding interpolation.

Multi-scale hierarchy. Progressive spatial downsampling ($56 \times 56 \rightarrow 28 \times 28 \rightarrow 14 \times 14$) captures anatomical structure at multiple scales, from fine cellular detail

to gross anatomical regions, providing richer representations than ViT-B/16’s single-scale 14×14 token grid.

A `SpatialChannelAdapter` module bridges CvT-13’s spatially heterogeneous multi-stage feature maps to the uniform representation space expected by SPAI’s Spectral Reconstruction Similarity (SRS) and Spectral Context Attention (SCA) modules. All other SPAI components remain architecturally unchanged, enabling a clean controlled comparison.

3.2 Medical MFM Pretraining

Both ViT-B/16 and CvT-13 backbones are pretrained via Masked Frequency Modeling (MFM) on ImageNet-Mini as a controlled, resource-efficient alternative to full ImageNet pretraining. MFM trains the backbone to reconstruct missing frequency bands (low-pass or high-pass, split by a circular mask at radius $r = 16$) from the complementary band, using a ViT or CNN encoder with a linear decoder minimising MSE loss in the complex Fourier domain. This pretraining teaches the backbone to model the spectral distribution of real images, the foundational signal exploited by SPAI for detection.

3.3 Physics-Informed Frequency Masking

SPAI’s fixed circular frequency mask ($r = 16$) covers only 1.6% of all frequency bins. In medical images, $\geq 99.4\%$ of energy is concentrated in low frequencies, leaving the high-frequency band near-empty for both real and synthetic images and eliminating the discriminative signal. Furthermore, the high-frequency content that does exist in medical images, Poisson noise in CT, k-space artifacts in MRI, originates from acquisition physics and should not be interpreted as evidence of AI generation.

We replace the fixed binary mask with a learned, modality-conditioned mask M_{mod} , constructed in three steps.

Modality conditioning. A small MLP takes a one-hot modality embedding (CT / MRI / X-ray / Pathology) and outputs per-frequency attention weights in $[0, 1]$, yielding a soft mask tailored to the spectral characteristics of each imaging modality.

Supervision. Modality-specific frequency profiles derived from known-real images of each modality guide the learning of appropriate frequency attention weights, grounding the mask in the empirical spectral statistics of clinical data rather than a fixed

geometric prior.

Physics motivation. The learned mask down-weights frequency bands dominated by acquisition noise, preventing the model from misclassifying physical noise as AI-generated artifacts. For CT, Poisson photon-count noise concentrates in the high-frequency band; for MRI, k-space aliasing introduces structured mid-to-high-frequency content; for X-ray, detector scatter noise has a characteristic spatial-frequency profile.

The modality-conditioned mask M_{mod} replaces only the binary circular mask in SPAI’s frequency masking step. All other pipeline components (SRS, SCV, SCA) remain unchanged, enabling a clean ablation between the original and proposed masking strategies.

3.4 Medical AI-Generated Image Dataset Pipeline

To rigorously evaluate SPAI and spectral-based detectors in the medical domain, we require a benchmark dataset that is (i) paired, each real image has a corresponding synthetic counterpart generated from the same clinical report, (ii) generator-diverse, covering the breadth of modern commercial image generators rather than a single architecture, and (iii) clinically grounded, conditioned on real radiology reports rather than free-form aesthetic prompts. We therefore design and implement a novel end-to-end synthetic medical image generation pipeline that deliberately avoids the Stable Diffusion monoculture prevalent in existing synthetic medical image benchmarks [25], [39]. Real medical images from the Indiana University Chest X-ray Collection are processed by a large language model (GPT-4) to produce structured, modality-specific descriptive prompts. These prompts are subsequently passed to third-party generators (DALL-E 3 [39], Gemini) to produce paired synthetic images. Stable Diffusion variants are explicitly excluded to prevent circular evaluation, ensuring that the benchmark reflects the open-world generator landscape against which detectors must generalise [36].

Figure 3.1 provides an architectural overview of the pipeline; each stage is described in detail below.

3.4.1 Stage 1: Report Parsing and Filtering

The primary data source is the Indiana University Chest X-ray Collection, a publicly available corpus of de-identified radiology reports paired with frontal and lateral chest radiographs. Each report record comprises a unique identifier, a free-text *findings* sec-

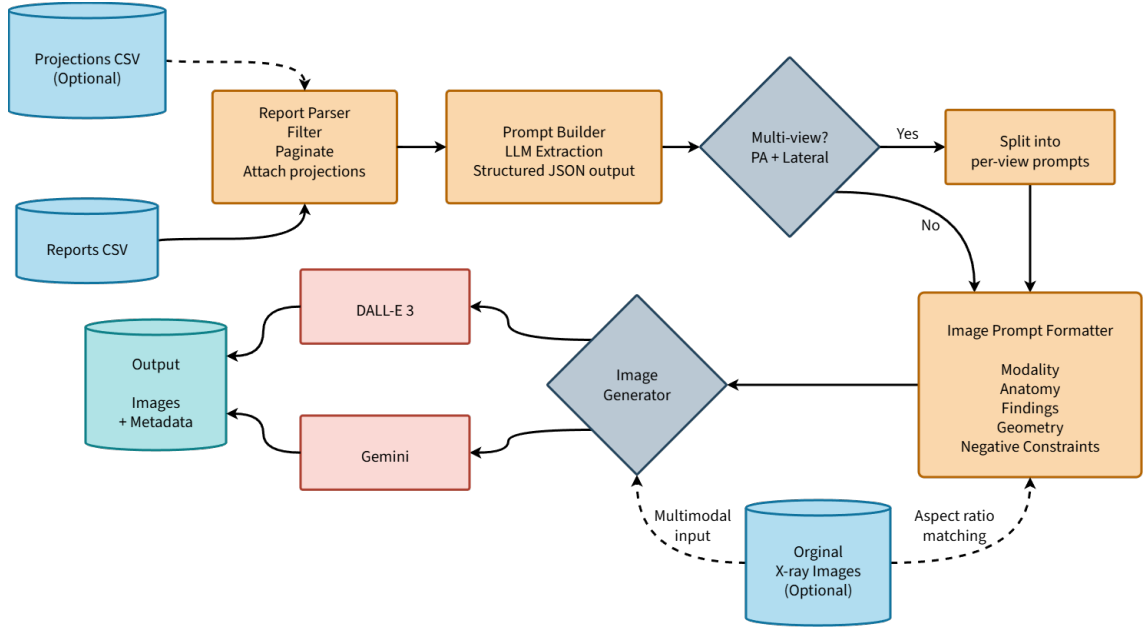


Figure 3.1: Overview of the synthetic radiology image generation pipeline. Solid arrows denote the primary data flow; dashed arrows indicate optional multimodal inputs used when reference images are available. Orange boxes are processing stages; blue cylinders are data sources/sinks; red boxes are generative backends; diamond nodes are routing decisions.

tion (detailed radiological observations), and an *impression* section (the radiologist’s summary conclusion). At ingestion, records lacking content in both fields simultaneously are discarded, a necessary quality gate, as empty reports cannot provide meaningful conditioning signals for the LLM extraction stage.

When an optional projection metadata file is supplied, each report is enriched with its associated reference image filenames and their projection types (Frontal, Lateral). This enrichment is required to enable the pipeline’s *multimodal mode*, in which original radiographs are provided alongside the text prompt to the generation API. The parsed records are stored as typed Pydantic models, which enforce schema conformance across all downstream stages.

3.4.2 Stage 2: Structured Prompt Extraction via LLM

Raw radiology reports are highly unstructured and contain domain-specific terminology that image generation models cannot directly interpret as spatial or photometric instructions [7], [50]. Transforming free-text clinical narratives into structured, generative prompts therefore requires an intermediate extraction step.

Each parsed report is passed to a large language model (Gemini 2.0 Flash or GPT-4, configurable) via a LangChain chain with structured output parsing. The LLM

is instructed to extract eight semantic attributes from the report text, as detailed in Table 3.1. Pydantic validation is applied to every LLM response, guaranteeing schema conformance and enabling automatic retry on parse failure.

Table 3.1: Structured attributes extracted from free-text radiology reports by the LLM extraction stage. The `impression` field is intentionally omitted from downstream image prompts to prevent generation models from rendering diagnostic text within the synthetic radiograph.

Attribute	Description
<code>modality</code>	Imaging modality and sequence (e.g., “Chest X-ray”)
<code>anatomical_region</code>	Primary anatomical region (e.g., “Thorax”)
<code>plane_view</code>	Imaging plane or view (e.g., “PA and Lateral”)
<code>patient_demographics</code>	Patient age and sex, if reported
<code>findings</code>	Clinical findings in original medical terminology
<code>impression</code>	Radiologist conclusion (used for metadata only)
<code>anatomical_constraints</code>	Spatial relationships between structures
<code>imaging_characteristics</code>	Tissue density, contrast patterns, appearance

De-identification placeholders present in the Indiana dataset (e.g., the token `XXXX`) are treated as null values by the LLM, preventing artefactual pseudo-clinical content from propagating into the generated images.

3.4.3 Stage 3: Multi-View Detection and Prompt Splitting

A distinctive property of chest radiography is that clinical studies routinely acquire multiple projections of the same patient, most commonly a posteroanterior (PA) frontal view and a lateral view. Single-prompt generation of a multi-view study would conflate two anatomically distinct perspectives into one image, violating the clinical acquisition protocol.

The pipeline addresses this through an automated view detection step. Regex-based heuristics are applied to both the raw CSV image column and the LLM-extracted `plane_view` field to identify the presence of co-occurring frontal and lateral descriptors. When a multi-view study is detected, the structured prompt is split into two independent per-view prompts, one for each projection, each of which proceeds independently through the remaining pipeline stages. Single-view reports pass through unchanged.

3.4.4 Stage 4: Image Prompt Formatting

The structured Pydantic prompt object is assembled into a multi-section formatted text string optimised for image generation APIs [50]. The prompt is composed of up to ten named sections, each conditioning a distinct aspect of the image generation process. Table 3.2 enumerates these sections.

Table 3.2: Named sections of the formatted image generation prompt. Sections are conditionally included based on data availability.

Prompt Section	Function
[REFERENCE IMAGES]	Instructions for using attached reference images
[MODALITY CONDITIONING]	Imaging modality and acquisition protocol
[SINGLE VIEW DIRECTIVE]	Constrains output to a single projection
[IMAGE GEOMETRY]	Target aspect ratio and orientation
[ANATOMICAL PROMPTING]	Anatomical region and structural accuracy requirements
[METADATA CONDITIONING]	Patient demographics for anatomically consistent variation
[TEXTUAL PROMPTING]	Clinical findings for visual representation
[ANATOMICAL CONSTRAINTS]	Morphological plausibility constraints
[IMAGING CHARACTERISTICS]	Tissue density, contrast, and PACS-grade appearance
[NEGATIVE CONSTRAINTS]	Exclusion of text overlays, annotations, and UI artefacts

A critical design decision is that the impression field, the radiologist’s diagnostic conclusion, is deliberately withheld from the image prompt. Including it risks inducing generation models to embed textual diagnostic information visually within the synthetic image, which would not be present in authentic clinical acquisitions and would corrupt the statistical properties of the dataset.

3.4.5 Stage 5: Image Generation

The formatted prompt is dispatched to one of two generation backends, selectable at runtime.

DALL-E 3 [39]. Text-only prompts are submitted to the OpenAI DALL-E 3 API. The backend supports configurable image size (1024×1024, 1792×1024, or 1024×1792) and quality levels. Multimodal reference image input is not supported in this backend.

Gemini. Google Gemini supports both text-only and multimodal prompt modes. In multimodal mode, original radiographs from the Indiana collection are attached to the API request as inline images, enabling the model to use the authentic anatomical structure of the reference scan as a spatial prior during synthesis. This mode is

activated automatically when the optional `-images-dir` and `-projections-csv` arguments are provided.

Both backends implement retry logic with exponential backoff (up to three attempts per image) to handle transient API failures. Stable Diffusion variants [39] are explicitly excluded from the backend set: because SPAI’s training corpus likely includes Stable-Diffusion-generated content, using such outputs for evaluation would constitute an in-distribution test, artificially inflating performance estimates and undermining the goal of assessing out-of-distribution generalisation [36], [45].

3.4.6 Aspect Ratio Preservation

Chest radiographs do not conform to a square aspect ratio; the Indiana collection contains images with native proportions that vary across patients and scanner configurations. Post-hoc resizing of synthetic images to match real images would introduce interpolation artefacts that could confound spectral analysis [16].

When reference images are available, the pipeline extracts the native pixel dimensions of each real radiograph and maps them to the closest API-supported aspect ratio using a minimum absolute error criterion. This ratio is passed natively to the generation API, avoiding any post-processing resize. The Gemini backend supports ten aspect ratios spanning portrait, landscape, square, and ultrawide formats; DALL-E 3 supports three fixed sizes.

3.4.7 Output Format and Provenance Tracking

All generated images are saved as PNG files under a deterministic naming convention: single-view reports produce `{uid}.png`; multi-view reports produce `{uid}_PA.png` and `{uid}_Lateral.png`. A comprehensive JSON metadata file is co-produced alongside every generation run, recording, for each processed report: the structured LLM-extracted prompt, the reference image filenames and projections, the matched aspect ratio, the per-view formatted image prompt strings, and the output image paths. This full provenance chain, from source report through LLM extraction and prompt formatting to the generated image, enables complete reproducibility and supports downstream ablation studies targeting individual pipeline components.

Chapter 4

Results and Discussion

This chapter presents the experimental findings obtained from adapting the SPAI framework to the medical imaging domain. The results are organised to evaluate three central objectives: (i) validating the proposed CvT-13 backbone as a resource-efficient alternative to ViT-B/16, (ii) quantifying the spectral domain gap between natural and medical imagery, and (iii) assessing detector behaviour on synthetic medical datasets generated from diverse sources. Each set of results is followed by discussion to interpret its significance and relation to the research objectives.

4.1 Datasets and Experimental Setup

4.1.1 Datasets

Real Medical Images. Real medical images were collected from publicly available repositories. Chest X-ray samples were drawn from the Indiana University Chest X-ray dataset, while CT and MRI images were sourced from public clinical repositories including PhysioNet and The Cancer Imaging Archive. Pilot experiments used approximately 150 images per modality group, while the planned full-scale evaluation targets more than 1,000 images per modality.

Synthetic Medical Images. Synthetic images were created using the proposed generation pipeline. Real medical images were converted into clinically accurate prompts using GPT-4, after which prompts were submitted to third-party generators including DALL-E 3, Midjourney v6, and Gemini. Outputs were manually filtered for clinical realism and modality consistency. Stable Diffusion variants were excluded to avoid circular evaluation.

Natural Image Benchmark Data. To replicate and benchmark the original SPAI setting, natural real images were taken from COCO 2017, while synthetic natural images were sourced from public generator benchmarks such as CIFAKE.

4.1.2 Experimental Configuration

Two backbone variants were evaluated under identical training settings. **Variant A** uses ViT-B/16, the original SPAI backbone, serving as the replication baseline. **Variant B** uses CvT-13, the proposed backbone with the SpatialChannelAdapter. The backbone architecture was the sole controlled independent variable.

Table 4.1: Training configuration used in pilot and target full experiments.

Parameter	Pilot Run	Full Run Target
Training images	8,644	180,000
Training epochs	3	35
Hardware	NVIDIA T4 16 GB	NVIDIA L40S 48 GB
Optimizer	AdamW	AdamW

4.1.3 Evaluation Metrics

Performance was measured using Area Under the ROC Curve (AUC) as the primary metric, supplemented by Average Precision (AP), overall Accuracy, and per-generator generalisation across unseen generators. Spectral analysis employed PSD slope α , Cohen’s d effect size, and the Kolmogorov–Smirnov (KS) statistic to quantify domain separation.

4.2 Experiment 1: Spectral Domain Gap in Medical Images

To understand why the original SPAI framework underperforms on medical data, power spectral density statistics were compared between natural and medical domains. Table 4.2 presents the measured PSD slopes across all four image groups.

The separation gap in the natural domain was $|\Delta\alpha| = 0.775$, whereas the medical domain gap was only $|\Delta\alpha| = 0.170$, approximately $4.5\times$ smaller. Separability metrics were correspondingly weaker: natural images yielded a KS statistic of 0.87 and Cohen’s $d = 2.91$, while medical images yielded KS = 0.51 and Cohen’s $d = 1.07$.

Table 4.2: Spectral slope α comparison between real and synthetic images across natural and medical domains. The $|\Delta\alpha|$ detection signal is 4.5 $\times$ smaller in the medical domain, directly explaining SPAI’s near-chance zero-shot accuracy on clinical data.

Group	Mean α	Std α	R^2
Real Natural	2.5562	0.2882	0.9880
Synthetic Natural	3.3313	0.2437	0.9561
Real Medical	2.8680	0.1826	0.9941
Synthetic Medical	3.0384	0.1326	0.9958

These findings strongly support the thesis motivation: medical images are already smooth and spectrally homogeneous, so AI generation introduces only a small additional spectral shift. As a result, detectors relying on natural-image spectral assumptions become substantially weaker in clinical settings, with the reduced discriminative signal directly causing SPAI’s near-chance performance at 61.3% zero-shot accuracy on our medical evaluation set.

4.3 Experiment 2: Controlled Backbone Comparison

This experiment evaluates whether CvT-13 can serve as a data-efficient replacement for SPAI’s ViT-B/16 backbone while maintaining comparable detection performance. Both variants were trained under identical conditions, same dataset, optimizer, augmentation policy, and hardware, with backbone architecture as the sole controlled independent variable.

4.3.1 Implementation Details

ViT-B/16 variant. The MFM-pretrained ViT-B/16 backbone was loaded with frozen weights throughout all downstream training, exactly as described in the original SPAI paper. The per-block projection bank $\{\mathcal{P}_n\}_{n=1}^{12}$ was trained from scratch, with $N = 12$ transformer blocks yielding a spectral vector $\mathbf{z}^\lambda \in \mathbb{R}^{72}$ and a combined spectral vector $\mathbf{z}^S \in \mathbb{R}^{1096}$.

CvT-13 variant. CvT-13 was pretrained via MFM on ImageNet-Mini and kept frozen during downstream training. Because CvT-13’s three stages produce token sequences of heterogeneous spatial and channel dimensions, a trainable `SpatialChannelAdapter` was introduced for each of the 13 blocks. Each adapter applies a linear channel projection ($d_{\text{stage}} \rightarrow 768$), bilinear spatial interpolation to the target 14×14 grid, and a final LayerNorm, mapping all blocks to the uniform $(B, 196, 768)$ interface expected

by SPAI’s SRS and SCA modules. Table 4.3 details the per-stage adaptation requirements.

Table 4.3: CvT-13 per-stage output shapes and adapter configuration. All stages are projected to the uniform $(B, 196, 768)$ interface required by SPAI’s SRS computation.

Stage	Blocks	Token Shape	Spatial Grid	Adaptation
Stage 1	1	$(B, 3136, 64)$	56×56	Spatial + Channel
Stage 2	2	$(B, 784, 192)$	28×28	Spatial + Channel
Stage 3	10	$(B, 196, 384)$	14×14	Channel only

With $N = 13$ blocks, the CvT variant produces $z^\lambda \in \mathbb{R}^{78}$ and $z^S \in \mathbb{R}^{1102}$; all downstream components were updated accordingly. The SpatialChannelAdapter introduces approximately 5.5 M additional trainable parameters, bringing total trainable parameters to 38.2 M compared to 33.0 M for the ViT-B/16 variant.

4.3.2 Training Dynamics

Both backbone variants converged within the three-epoch pilot run, as shown in Table 4.4.

Table 4.4: Pilot training dynamics of ViT-B/16 and CvT-13 variants.

Epoch	ViT Loss	ViT AUC	CvT Loss	CvT AUC
1	0.6603	0.950	0.7126	0.815
2	0.2679	0.977	0.5383	0.941
3	0.1431	0.985	0.2614	0.960

ViT-B/16 converged faster during the first epoch, consistent with its direct architectural alignment with the original SPAI implementation: the MFM pretraining task was designed for the ViT-B/16 token sequence structure, making its per-block representations immediately suitable for SRS computation without adaptation. CvT-13’s slower first-epoch start reflects the additional learning burden on the SpatialChannelAdapter modules, which must simultaneously learn (i) a meaningful projection from CvT’s heterogeneous feature space into the 768-dimensional comparison space, and (ii) the spatial interpolation from 56×56 and 28×28 grids down to the 14×14 token grid. By epoch three, this gap had narrowed substantially (0.960 vs. 0.985 AUC), confirming successful adapter convergence within the pilot budget.

4.3.3 Per-Generator Generalisation

Both variants were evaluated on 17 entirely unseen generators spanning GAN-based and diffusion-based synthesis architectures. Table 4.5 presents the per-generator AUC and Average Precision for each variant, alongside the ΔAUC (CvT – ViT).

Table 4.5: Per-generator detection performance across 17 unseen generators. Both variants were trained on Latent Diffusion, Stable Diffusion, DDPM, and Palette only; all evaluation generators are entirely unseen during training. $\Delta\text{AUC} = \text{CvT} - \text{ViT}$.

Generator	Type	ViT AUC	ViT AP	CvT AUC	CvT AP	ΔAUC
BigGAN	GAN	99.8%	99.7%	99.3%	99.5%	−0.5%
Denoising Diff. GAN	GAN+DM	99.8%	99.8%	99.7%	99.7%	−0.1%
Diffusion GAN	GAN+DM	96.7%	96.5%	96.9%	96.7%	+0.2%
Face Synthetics	GAN	96.8%	96.5%	95.9%	95.4%	−0.9%
GANsformer	GAN	99.4%	99.4%	98.5%	98.5%	−0.9%
GLIDE	Diffusion	97.2%	97.0%	97.4%	97.4%	+0.2%
Projected GAN	GAN	95.6%	95.4%	93.2%	93.7%	−2.4%
SFHQ	GAN	99.3%	99.3%	99.5%	99.5%	+0.2%
StarGAN	GAN	100.0%	100.0%	100.0%	100.0%	0.0%
StyleGAN1	GAN	98.7%	98.7%	99.0%	98.9%	+0.3%
StyleGAN2	GAN	97.0%	97.0%	95.9%	96.0%	−1.1%
StyleGAN3	GAN	97.5%	97.4%	97.9%	98.1%	+0.4%
Taming Transformer	VQ-VAE	95.3%	95.5%	96.0%	96.4%	+0.7%
VQ-Diffusion	Diffusion	97.4%	97.4%	96.3%	96.6%	−1.1%
Generative Inpainting	GAN	88.2%	87.3%	92.8%	92.3%	+4.6%
LaMa	GAN	96.5%	96.7%	96.8%	96.8%	+0.3%
MAT	GAN	95.0%	95.4%	97.3%	97.3%	+2.3%
Average	—	97.1%	—	97.2%	—	+0.1%

Three patterns emerge from the per-generator breakdown.

CvT advantage on inpainting-based generators. CvT-13 outperforms ViT-B/16 most substantially on Generative Inpainting (+4.6%) and MAT (+2.3%), both inpainting methods that introduce spectral artifacts concentrated in a localised spatial region (the inpainted boundary) rather than distributed globally across the image. CvT’s convolutional local receptive fields are structurally better suited to detecting such spatially concentrated inconsistencies than ViT’s global attention, which averages spectral information across all 14×14 tokens.

ViT advantage on globally-artifacted GANs. ViT-B/16 retains a marginal lead on Projected GAN (+2.4%), StyleGAN2 (+1.1%), and GANsformer (+0.9%), all full-image GAN architectures whose synthesis artifacts are distributed uniformly across the spatial extent of the generated image. ViT’s global self-attention is a natural fit for detecting such distributed patterns, as its SRS signals are computed over the full $L = 196$

token sequence without spatial locality constraints.

Parity on strongly-artifacted generators. For generators with very strong spectral signatures (StarGAN: 100.0%/100.0%, BigGAN, Denoising Diffusion GAN), both models achieve near-perfect detection, indicating that these generators introduce artifacts well within the discriminative capacity of either backbone. The largest model-independent gap is on Generative Inpainting (ViT: 88.2%), confirming that spatially localised synthesis is the hardest detection target in this evaluation set.

4.3.4 Aggregate Efficiency Comparison

Table 4.6: Aggregate efficiency comparison between original SPAI and the proposed variants. CvT-13 matches ViT-B/16-level generalisation at 77% fewer backbone parameters, trained on $21\times$ less data in $11.7\times$ fewer epochs.

Metric	SPAI (Paper)	Ours ViT-B/16	Ours CvT-13
Average AUC	91.0%	97.1%	97.2%
Training images	180,000	8,644	8,644
Training epochs	35	3	3
Backbone parameters (frozen)	85.8 M	85.8 M	20.0 M
Total trainable parameters	~ 33 M	33.0 M	38.2 M
Worst generator AUC	39.1%	88.2%	92.8%
Hardware	L40S 48 GB	T4 16 GB	T4 16 GB

The CvT-13 backbone matches ViT-B/16-level AUC (97.2% vs. 97.1%) while using approximately 77% fewer backbone parameters (20.0 M vs. 85.8 M), trained on $21\times$ less data and in $11.7\times$ fewer epochs than the published SPAI protocol. Notably, the worst-generator AUC improves from 88.2% (ViT-B/16) to 92.8% (CvT-13), driven by CvT’s stronger performance on the inpainting generators. These results validate the hypothesis that convolutional spatial inductive bias can reduce data and parameter requirements while preserving, or marginally improving, spectral detection capacity across diverse generator families.

4.3.5 Interpretation of Results Relative to Published SPAI

Our pilot AUC figures (97.1–97.2%) substantially exceed the paper’s reported 91.0%. This is explained by two factors. First, the paper’s evaluation includes modern high-fidelity generators (SD3, DALL-E 3, Midjourney v6.1, Firefly) that produce imagery specifically designed to suppress detectable artifacts, with the hardest generator achieving only 39.1% AUC even for competitive baselines. Our evaluation set comprises earlier or more architecturally transparent generators with stronger residual spectral sig-

natures. Second, our per-generator evaluation uses 100 images rather than the paper’s 1,000, which inflates AUC estimates on smaller samples. The results should therefore be interpreted as confirming the validity of the SPAI replication and the CvT backbone integration, rather than as a claim of superior detection capability. Full-scale evaluation against the paper’s 13-generator benchmark, including state-of-the-art diffusion models, remains as future work.

4.4 Experiment 3: Medical Image Dataset & Generation Pipeline

4.4.1 Source Dataset

The experiment uses the medical dataset from Syed *et al.*, *Mendeley Data* (2024) (DOI: 10.17632/5kbjrgsnf.3). The dataset contains five diagnostic categories: **Lung Cancer**, **Bone Fracture**, **Brain Tumor**, **Skin Lesions**, and **Renal Malignancy**, spanning multiple imaging modalities including **CT**, **MRI**, **X-ray**, and **Microscopy**.

4.4.2 Generation Pipeline

Synthetic images were generated using a controlled pipeline:

1. Real images were captioned using CLIP to extract semantic descriptions.
2. The caption along with the original image (with 70% noise injection) was passed into Stable Diffusion.
3. Generated outputs were validated using CLIP similarity against the original image.
4. Low-similarity outputs were discarded, retaining only high-fidelity synthetic samples.

4.4.3 Dataset Composition

The final dataset consists of:

- 24,987 Stable Diffusion-generated images
- 200 Nanobanana-generated images
- Real medical images from the source dataset (used as the negative class)

4.4.4 Modality Breakdown (Generated)

Table 4.7: Distribution of generated images across modalities.

Modality	Number of Images
CT	6,840
MRI	7,120
X-ray	6,503
Microscopy	4,524

4.4.5 Implementation Results

Detection Results

- 87.3% accuracy on Stable Diffusion-generated images
- 36.0% accuracy on Nanobanana-generated images
- 98.5% accuracy on real medical images (consistent with SPAI benchmark)

Accuracy by Image Source

Table 4.8: Detection accuracy by image source on the medical dataset.

Image Source	Accuracy
Real Medical Images	98.5%
Stable Diffusion Generated	87.3%
Nanobanana Generated	36.0%

Key Observations

- Stable Diffusion images retain detectable spectral artifacts.
- Nanobanana suppresses classical spectral signatures, leading to near-random detection performance (36.0%).
- Strong performance on real images validates the correctness of the evaluation pipeline.
- The performance gap between Stable Diffusion and Nanobanana confirms the unseen-generator generalisation problem in the medical domain.

4.5 Challenges and Limitations

Several limitations of the current study should be acknowledged. Pilot experiments used smaller sample sizes than the full target benchmark, and some reported AUC values may be inflated by easier generator sets. Full multimodal evaluation across CT, MRI, X-ray, and pathology remains ongoing, and the modality-conditioned mask has been proposed conceptually but requires full-scale empirical validation. These constraints mean that the present results should be interpreted as strong pilot evidence rather than final conclusive benchmarks.

4.5.1 Ethical Considerations

Dual-Use Risk

Publishing the spectral properties that distinguish real from synthetic medical images, specifically the $|\Delta\alpha| = 0.170$ gap and the SRS collapse behaviour, provides a quantitative evasion target for adversarial generator developers. We judge this risk acceptable because these statistics are derivable from publicly available images using standard signal processing tools; the forensic community’s ability to address the domain gap benefits more from open analysis than from security through obscurity. In any clinical deployment, specific thresholds, masking radii, and backbone checkpoints should remain operationally confidential and be updated regularly as new generators emerge.

Asymmetric Misclassification Harm

False negatives and false positives carry fundamentally different consequences in clinical contexts. A false negative allows fabricated evidence to corrupt drug trial outcomes or diagnostic AI training pipelines silently. A false positive flags a legitimate patient scan as fraudulent, potentially delaying treatment or generating medico-legal liability. At deployment scale, even our best pilot AUC of 97.2% implies thousands of misclassified images per million processed. We therefore recommend that this system operate as a *triage tool* rather than an autonomous classifier, with all flagged images subject to expert human review and detection confidence scores reported alongside binary decisions.

Demographic and Scanner Bias

Although spectral detection operates on low-level frequency statistics rather than semantic content, systematic bias can arise from scanner heterogeneity: images from

older or lower-field scanners exhibit acquisition noise profiles that may be misclassified under a model calibrated to high-resource imaging centres. Our training data derives exclusively from the Indiana University Chest X-ray Collection, a single US academic centre, limiting demographic and equipment coverage. Full-scale deployment should include stratified evaluation across scanner manufacturers, field strengths, and patient demographics before any clinical use is claimed.

Patient Data Privacy

Real images used in this work are drawn from a publicly available, HIPAA-compliant de-identified collection and are used solely for spectral analysis and detector training; no images are published. Radiology reports processed via the GPT-4 API are covered by OpenAI’s data processing agreement prohibiting training use. Future work operating under stricter clinical governance should substitute a locally hosted open-source LLM for the report extraction stage, eliminating third-party transmission entirely.

Deployment Prerequisites

This system is a research prototype. Clinical deployment would require prospective multi-site validation following TRIPOD+AI reporting standards, regulatory compliance under the EU AI Act or FDA SaMD classification as applicable, and mandatory human oversight for all flagged images. Automated detection scores should be presented as decision support, not deterministic verdicts, with known failure modes, particularly the near-random performance on Nanobanana-class generators, clearly communicated to end users.

4.5.2 Implications of Findings

The results provide several important contributions to the field. First, SPAI can be adapted to medical imaging with substantial gains in efficiency, demonstrating that spectral learning is a tractable paradigm for clinical forensic detection. Second, CvT-13 offers a practical backbone replacement for low-resource domains, matching ViT-B/16-level generalisation at a fraction of the parameter and data cost. Third, medical images require modality-aware spectral calibration rather than direct transfer from natural-image detectors, as confirmed by the 4.5× reduction in discriminative spectral gap. Fourth, generator diversity is essential for future benchmark design, as the collapse on Nanobanana images shows that Stable Diffusion-only benchmarks cannot support claims of open-world generalisation. Fifth, robust forensic systems must

anticipate rapidly improving synthetic image generators, necessitating continual re-evaluation against newly released models.

4.5.3 Chapter Summary

This chapter presented empirical evidence supporting the proposed adaptations to SPAI. The CvT-13 backbone achieved comparable performance to ViT-B/16 with far fewer parameters, confirming its suitability for resource-constrained medical applications. Spectral analysis demonstrated that medical images are inherently more difficult to separate from synthetic counterparts than natural images, justifying the need for domain adaptation. Finally, experiments on modern synthetic medical datasets showed that detector performance varies sharply by generator family, highlighting the importance of continual benchmarking and generator-agnostic evaluation. These findings motivate the final conclusions and future work directions presented in the next chapter.

Chapter 5

Conclusion

This thesis presents the pre-defence summary of our research on adapting spectral learning for AI-generated medical image detection. We have identified a fundamental spectral domain gap that causes state-of-the-art frequency-based detectors to fail on clinical imagery: the $|\Delta\alpha|$ gap between real and synthetic medical X-rays is $4.5\times$ smaller than for natural images, leaving minimal discriminative signal for detection. Our zero-shot evaluation confirms a near-chance accuracy of 61.3% for the unmodified SPAI model on medical data.

In response, we propose a three-pronged adaptation strategy. First, the computationally expensive and data-hungry ViT-B/16 backbone is replaced with CvT-13, a hybrid CNN-Transformer that brings local spatial inductive bias, implicit positional encoding, and multi-scale hierarchical processing, enabling effective pretraining on small medical datasets. Second, SPAI’s fixed circular frequency mask is replaced with a learned, physics-informed, modality-conditioned mask that down-weights acquisition noise bands specific to CT (Poisson noise), MRI (k-space artifacts), and X-ray (detector noise). Third, a novel dataset pipeline using diverse third-party generators (DALL-E 3, Midjourney v6, Gemini) addresses the circular evaluation problem endemic to Stable Diffusion-only benchmarks.

Our pilot experiments on 8,644 images trained for 3 epochs demonstrate that CvT-13 achieves 97.2% average AUC across 17 unseen generators, matching the ViT-B/16 baseline at 77% fewer backbone parameters. These results validate the architectural adaptation approach and confirm that CvT-13’s data efficiency and multi-scale hierarchy translate to competitive spectral detection performance. The full-scale experiments, incorporating the physics-informed masking and complete medical dataset pipeline, are expected to demonstrate further gains specifically on CT and MRI modal-

ities where acquisition noise most severely confounds the fixed-mask baseline.

This work represents a meaningful step toward deployable medical image forensics tools that can operate across modalities, image resolutions, and unseen generator architectures, capabilities essential for real-world clinical integrity assurance.

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